



US012414201B2

(12) **United States Patent**
Sun et al.

(10) **Patent No.:** **US 12,414,201 B2**
(45) **Date of Patent:** ***Sep. 9, 2025**

(54) **5G NR FR2 BEAM MANAGEMENT
ENHANCEMENTS**

(71) Applicant: **Apple Inc.**, Cupertino, CA (US)

(72) Inventors: **Haitong Sun**, Irvine, CA (US); **Dawei Zhang**, Saratoga, CA (US); **Hong He**, Cupertino, CA (US); **Jia Tang**, San Jose, CA (US); **Pengkai Zhao**, San Jose, CA (US); **Tianyan Pu**, Cupertino, CA (US); **Wei Zeng**, San Diego, CA (US); **Wei Zhang**, Santa Clara, CA (US); **Yuchul Kim**, Santa Clara, CA (US); **Yushu Zhang**, Beijing (CN)

(73) Assignee: **Apple Inc.**, Cupertino, CA (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **18/507,156**

(22) Filed: **Nov. 13, 2023**

(65) **Prior Publication Data**

US 2024/0090081 A1 Mar. 14, 2024

Related U.S. Application Data

(63) Continuation of application No. 17/731,995, filed on Apr. 28, 2022, now Pat. No. 11,864,278, which is a
(Continued)

(51) **Int. Cl.**
H04W 88/06 (2009.01)
H04B 7/06 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **H04W 88/06** (2013.01); **H04B 7/0626** (2013.01); **H04B 7/0695** (2013.01);
(Continued)

(58) **Field of Classification Search**

CPC H04W 88/06; H04W 52/24; H04W 72/21;
H04W 76/11; H04W 76/27; H04W 80/02;
(Continued)

(56) **References Cited**

U.S. PATENT DOCUMENTS

10,966,183 B2 3/2021 Wang
11,224,021 B2* 1/2022 Cirik H04W 52/146
(Continued)

FOREIGN PATENT DOCUMENTS

CN 110034876 7/2019
CN 110463264 11/2019
(Continued)

OTHER PUBLICATIONS

Partial European Search Report for EP Patent Application No. 20207864.8-1216; Mar. 30, 2021.
(Continued)

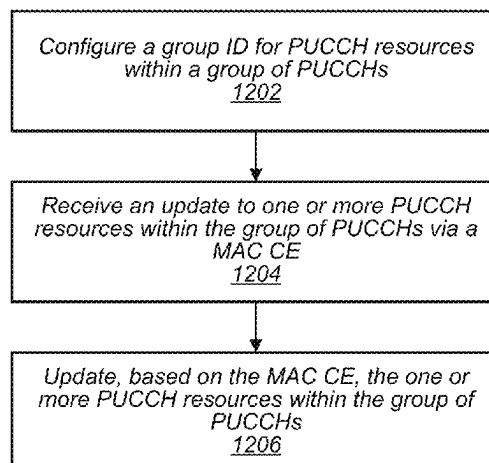
Primary Examiner — Hassan A Phillips

Assistant Examiner — Prenell P Jones

(74) *Attorney, Agent, or Firm* — Kowert, Hood, Munyon, Rankin & Goetzel, P.C.

(57) **ABSTRACT**

Apparatuses, systems, and methods for a user equipment device (UE) to perform methods for Fifth Generation New Radio (5G NR) frequency range two (FR2) based communications beam management. The UE may configure, via communications with a base station, serving the UE, a group ID for PUCCH resources within a group of PUCCHs, e.g., for FR2 based communications. The UE may receive, from the base station, an update to one or more PUCCH resources within the group of PUCCHs via a medium access control (MAC) control element (CE) that may include one of a resource ID indicating an update to a PUCCH resource or the group ID to update the group of PUCCH resources. The
(Continued)



UE may update, based on the MAC-CE, the one or more PUCCH resources within the group of PUCCHs.

20 Claims, 15 Drawing Sheets

Related U.S. Application Data

continuation of application No. 17/021,136, filed on Sep. 15, 2020, now Pat. No. 11,330,668.

(60) Provisional application No. 62/939,215, filed on Nov. 22, 2019.

(51) Int. Cl.

H04L 5/00 (2006.01)

H04L 25/02 (2006.01)

H04W 52/24 (2009.01)

H04W 56/00 (2009.01)

H04W 72/21 (2023.01)

H04W 76/11 (2018.01)

H04W 76/27 (2018.01)

H04W 80/02 (2009.01)

(52) U.S. Cl.

CPC **H04L 5/0051** (2013.01); **H04L 25/0226** (2013.01); **H04W 52/242** (2013.01); **H04W 56/001** (2013.01); **H04W 72/21** (2023.01); **H04W 76/11** (2018.02); **H04W 76/27** (2018.02); **H04W 80/02** (2013.01)

(58) Field of Classification Search

CPC H04W 72/044; H04W 72/20; H04B 7/06; H04B 7/08; H04B 7/0408; H04L 5/00; H04L 25/02

See application file for complete search history.

(56) References Cited

U.S. PATENT DOCUMENTS

11,510,239 B2 *	11/2022	Cirik	H04W 74/002
2008/0076359 A1	3/2008	Charpentier	
2008/0139113 A1	6/2008	Ho	
2008/0212561 A1	9/2008	Pani	
2009/0040982 A1	2/2009	Ho	
2013/0279490 A1	10/2013	Pani	
2016/0323873 A1	11/2016	Takeda	
2017/0171060 A1	6/2017	Liu	
2017/0331577 A1	11/2017	Parkvall	

2019/0132033 A1 *	5/2019	Akkarakaran	H04W 52/42
2019/0149379 A1	5/2019	Xiong	
2019/0281588 A1 *	9/2019	Zhang	H04W 52/08
2019/0312698 A1	10/2019	Akkarakaran	
2019/0357092 A1	11/2019	Jung	
2020/0100232 A1	3/2020	Onggosanusi	
2020/0177266 A1	6/2020	Kang	
2020/0205085 A1 *	6/2020	Li	H04W 52/242
2020/0213978 A1	7/2020	Iyer	
2020/0267668 A1 *	8/2020	Xu	H04W 72/0473
2020/0313747 A1	10/2020	Xu	
2020/0314829 A1	10/2020	Venugopal	
2020/0314860 A1 *	10/2020	Zhou	H04W 72/23
2020/0351129 A1 *	11/2020	Kwak	H04W 76/27
2020/0351905 A1	11/2020	Venugopal	
2020/0389210 A1	12/2020	Grant	
2021/0029649 A1 *	1/2021	Cirik	H04W 52/0212
2021/0045070 A1	2/2021	Yi	
2021/0051508 A1	2/2021	Chung	
2021/0051653 A1	2/2021	Park	
2021/0051672 A1	2/2021	Rastegardoost	
2021/0105726 A1	4/2021	Yao	
2021/0105753 A1	4/2021	Zhang	
2021/0135741 A1 *	5/2021	Zhou	H04B 7/088
2021/0143922 A1 *	5/2021	Jin	H04W 24/08
2021/0144742 A1	5/2021	Jeon	
2021/0153040 A1	5/2021	Zhou	
2021/0153137 A1	5/2021	Molavianjazi	
2021/0160964 A1 *	5/2021	Sun	H04B 7/0626
2021/0168574 A1	6/2021	Zhang	
2021/0194639 A1	6/2021	Yu	
2021/0211988 A1	7/2021	Jung	
2021/0219196 A1	7/2021	Jeon	
2022/0225242 A1 *	7/2022	Guan	H04W 52/54
2022/0271890 A1 *	8/2022	Grossmann	H04L 5/0051
2022/0303789 A1 *	9/2022	He	H04B 7/0695
2022/0394626 A1 *	12/2022	Matsumura	H04W 72/04

FOREIGN PATENT DOCUMENTS

WO	2016171748	10/2016
WO	2019108340	6/2019
WO	2019190377	10/2019
WO	2020162716	8/2020

OTHER PUBLICATIONS

Lenovo et al. "Discussion of multi-beam operation"; 3GPP TSG RAN WG1 Meeting #99 R1-1912317; Nov. 18, 2019.
Office Action for CN Patent Application No. 202011306256.7; Mar. 1, 2024.

* cited by examiner

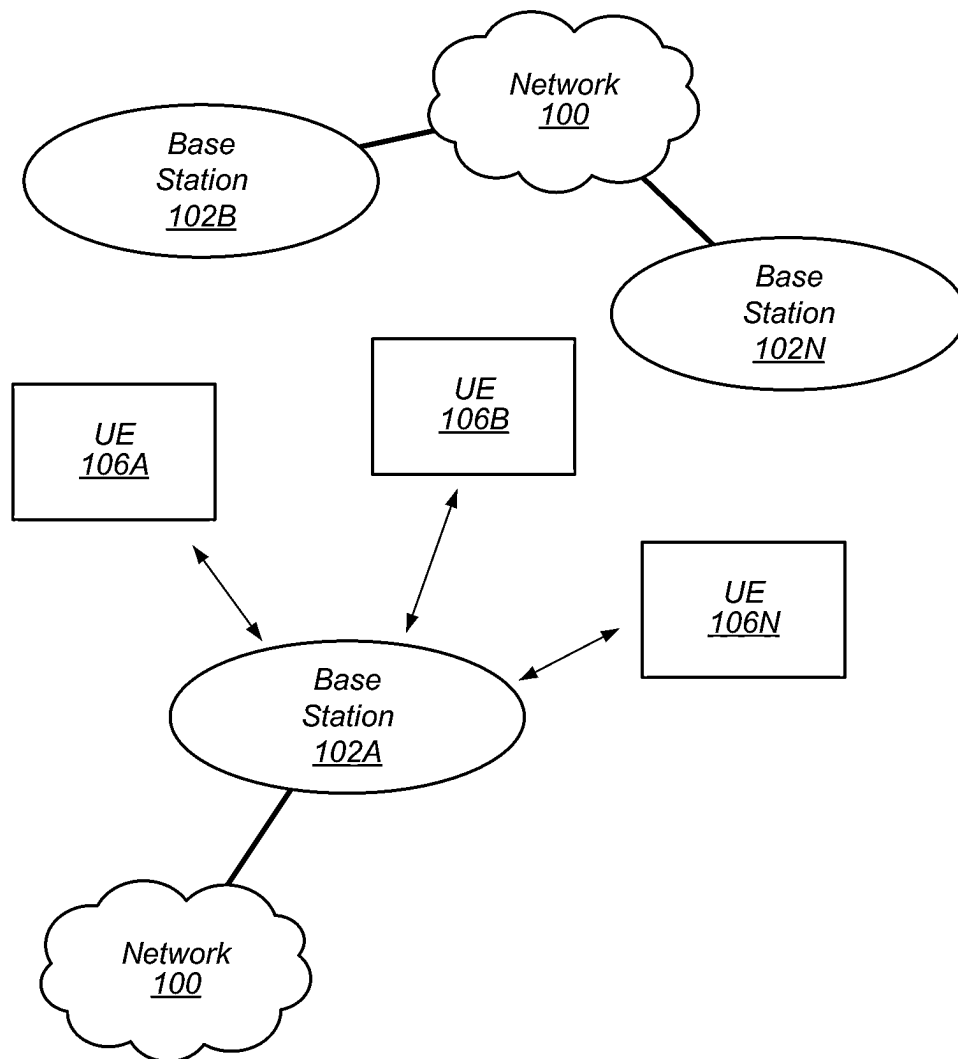


FIG. 1A

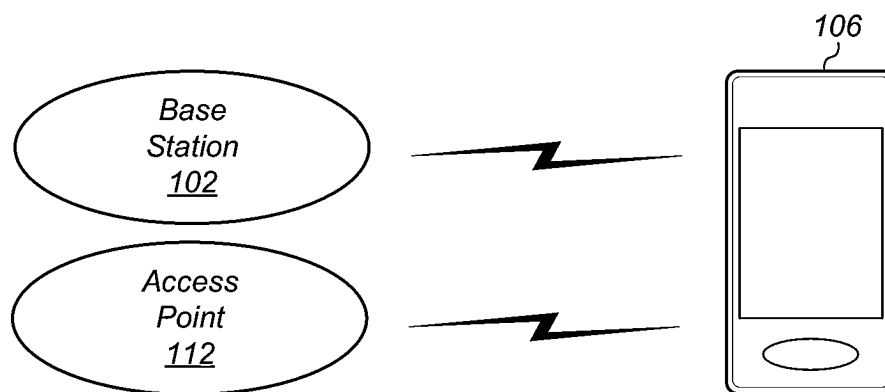


FIG. 1B

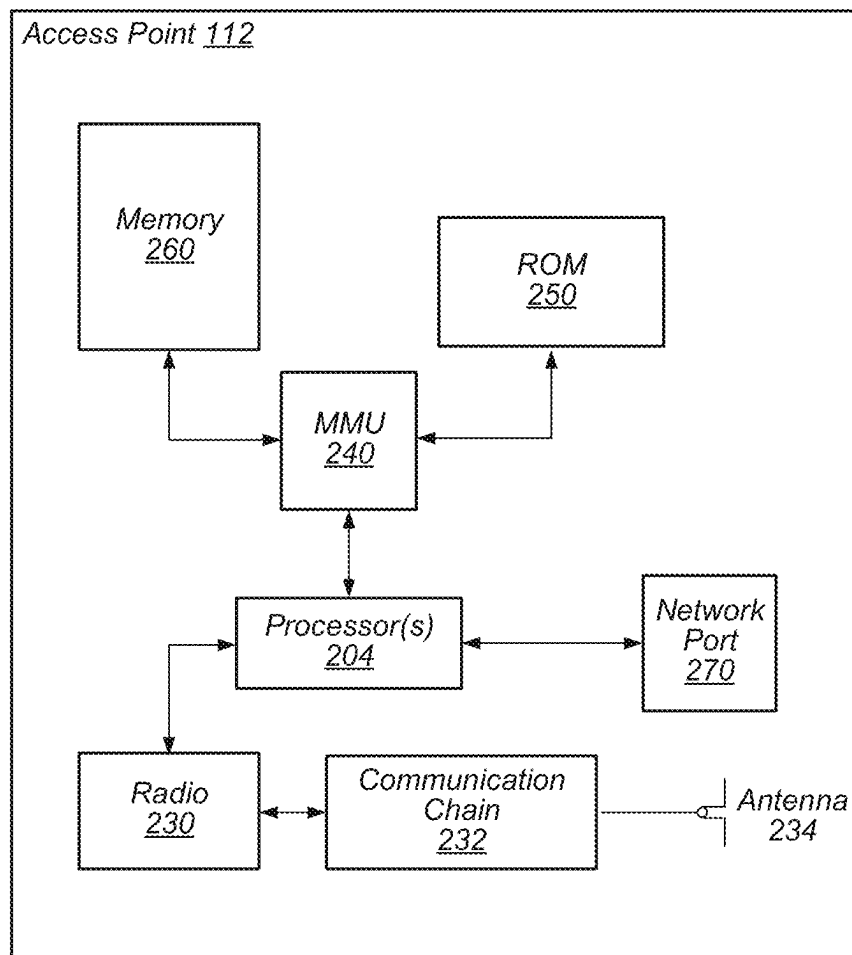


FIG. 2

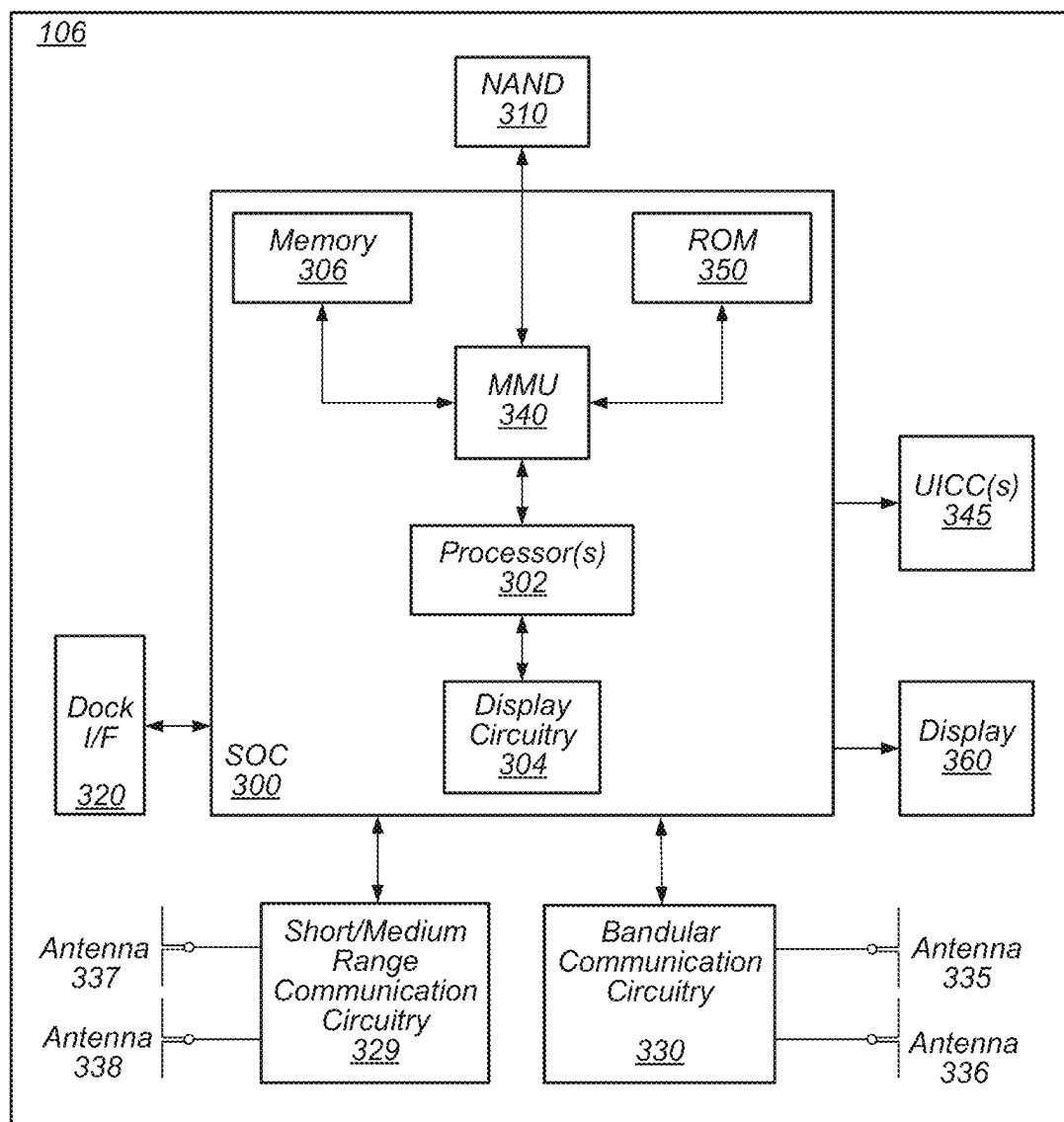


FIG. 3

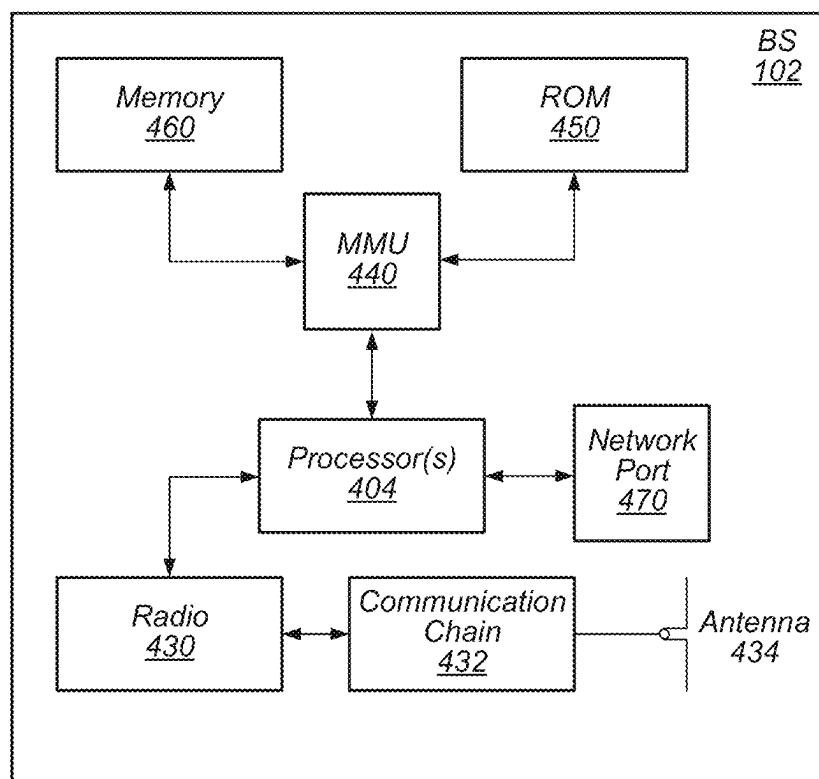


FIG. 4

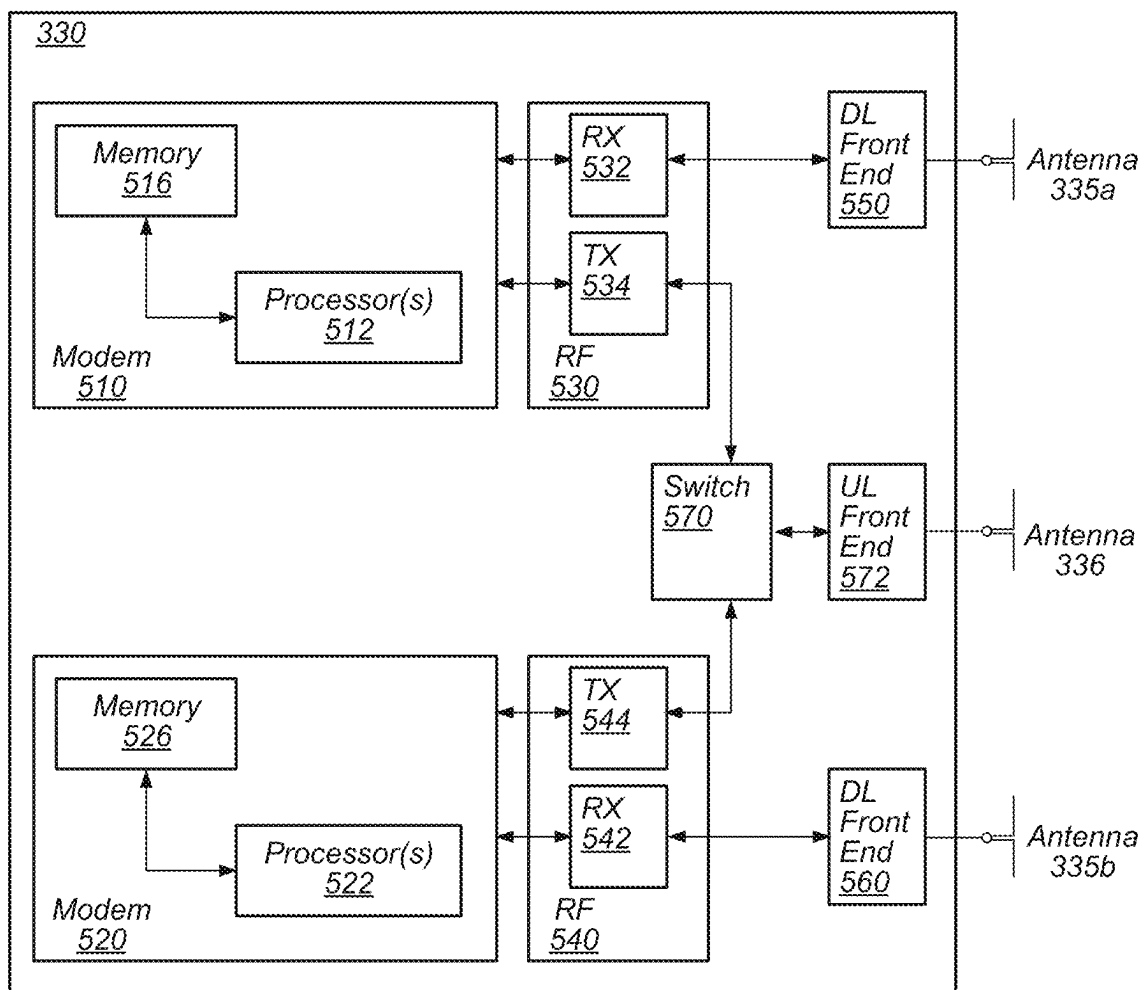


FIG. 5

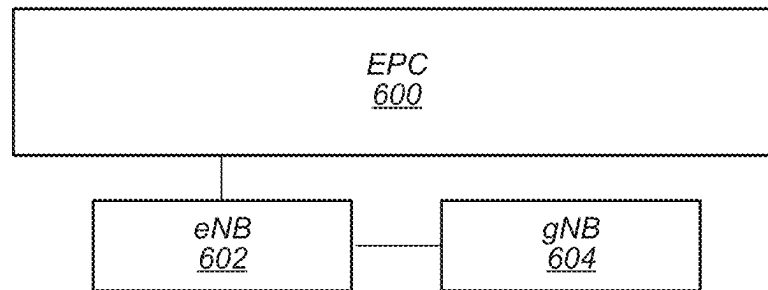


FIG. 6A

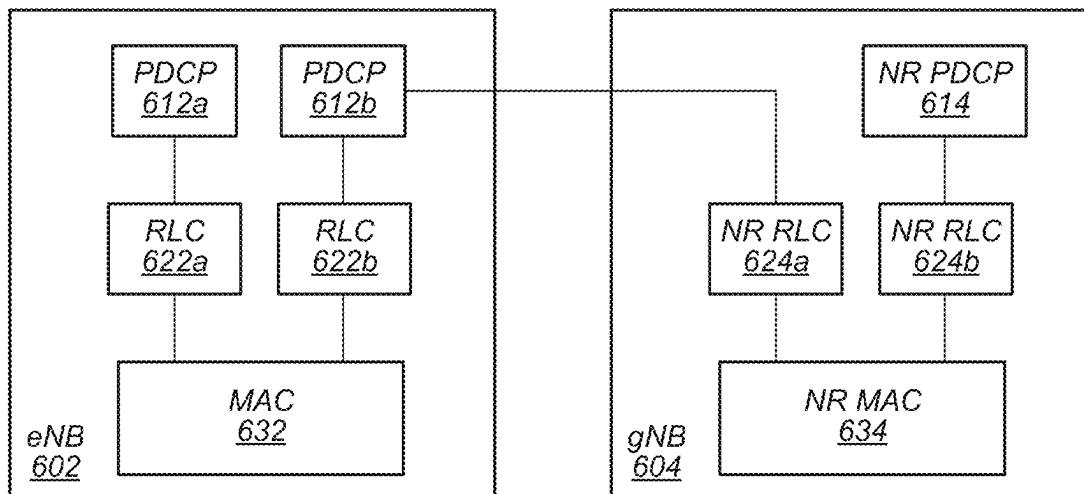
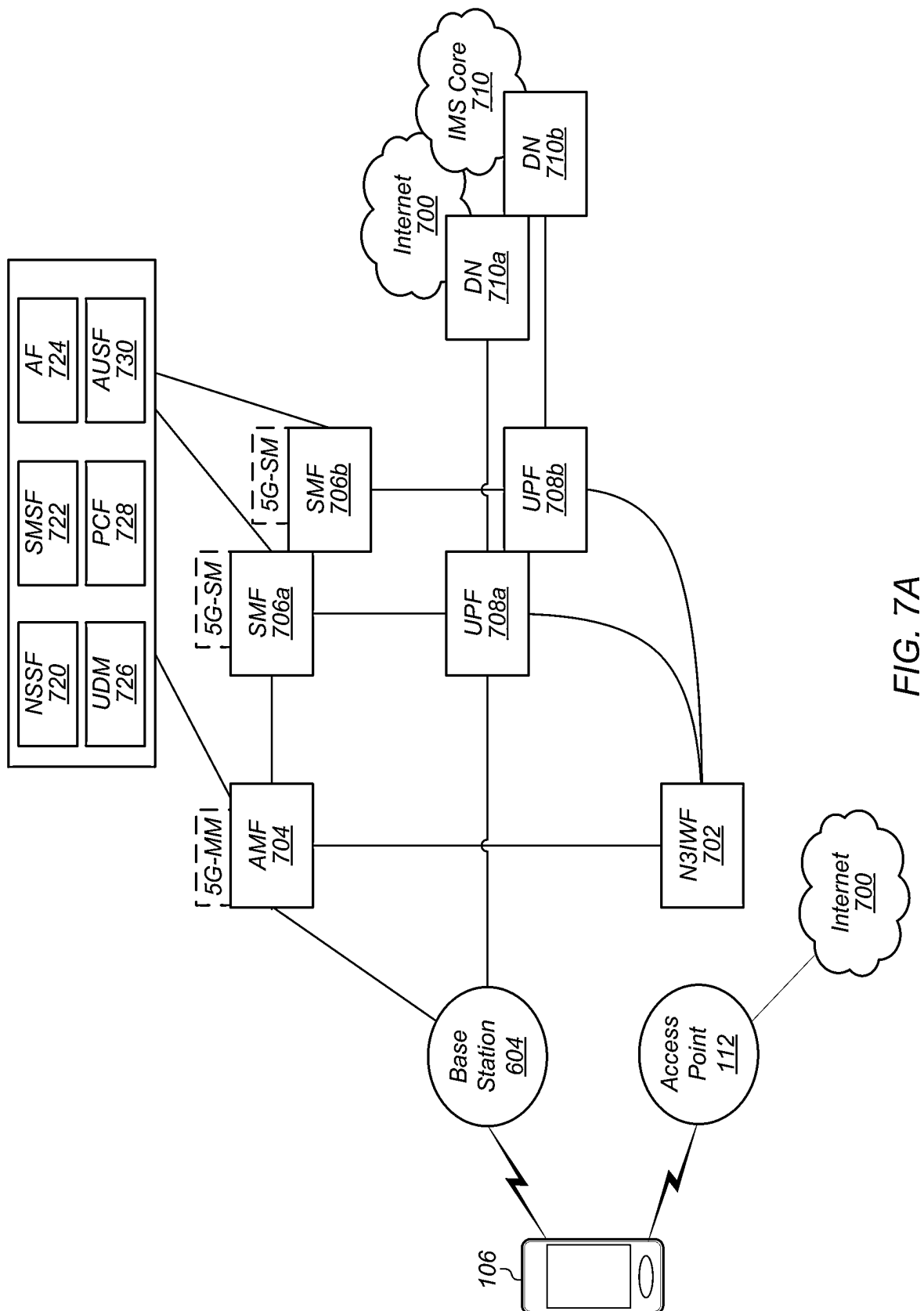


FIG. 6B



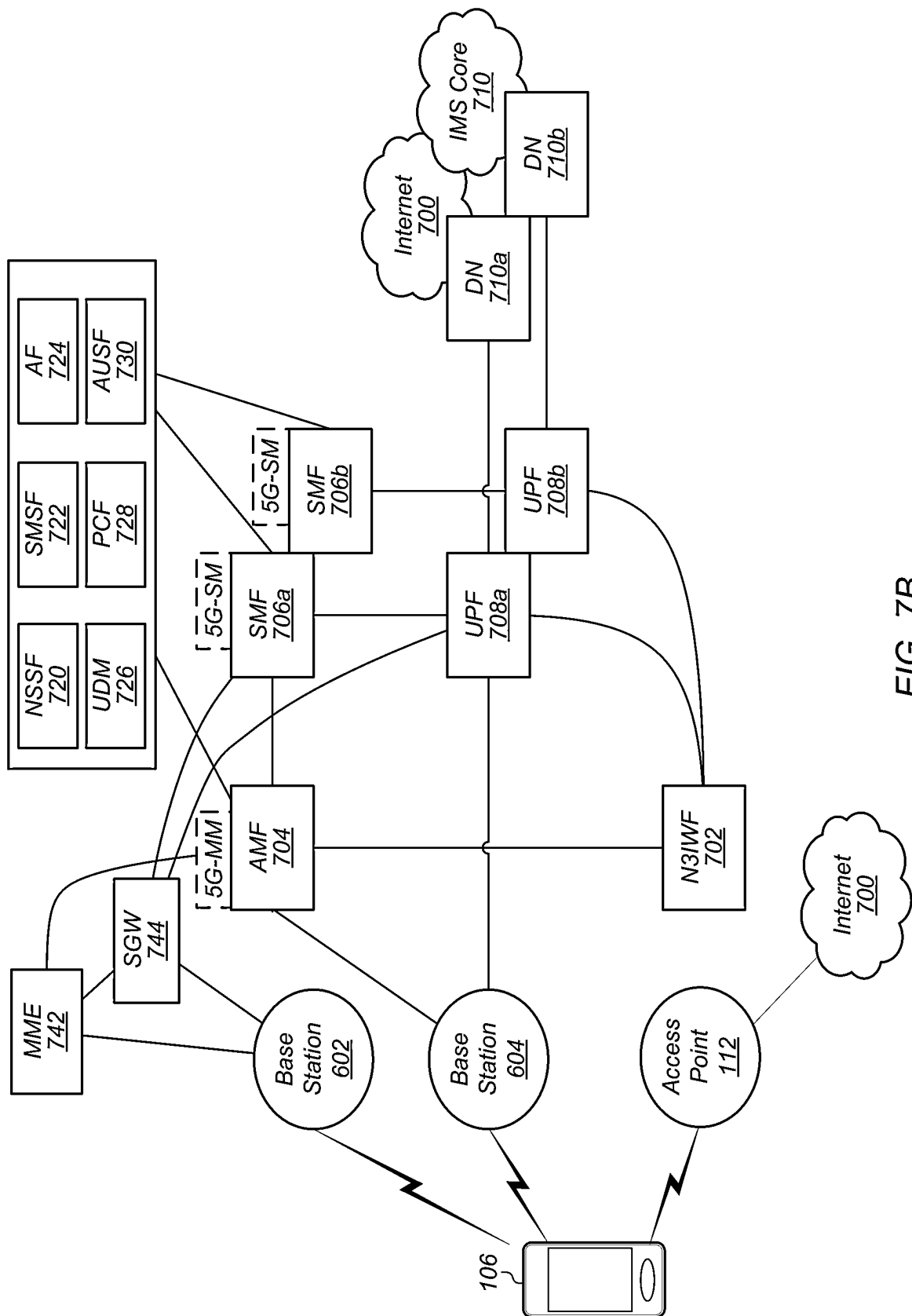


FIG. 7B

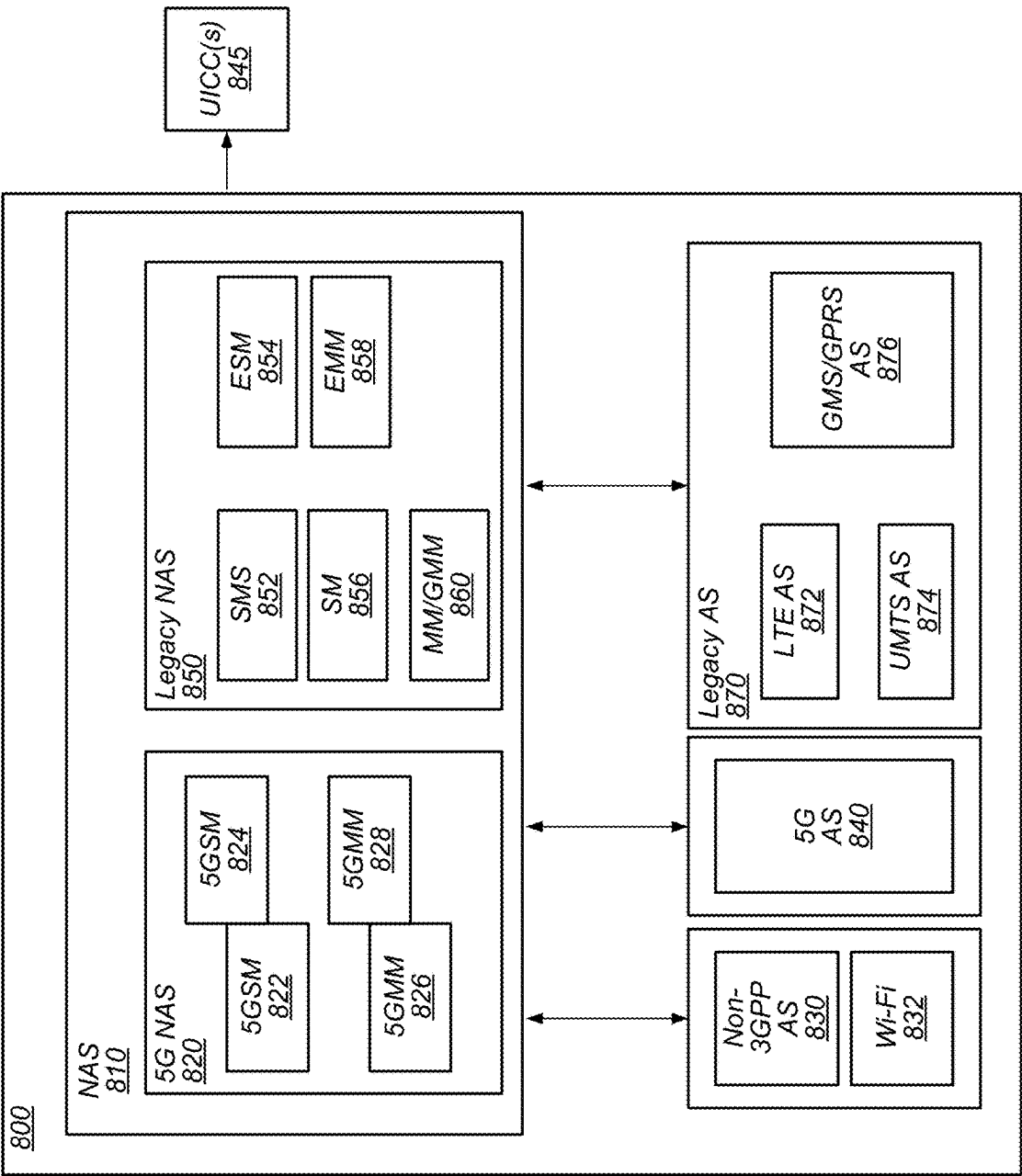


FIG. 8

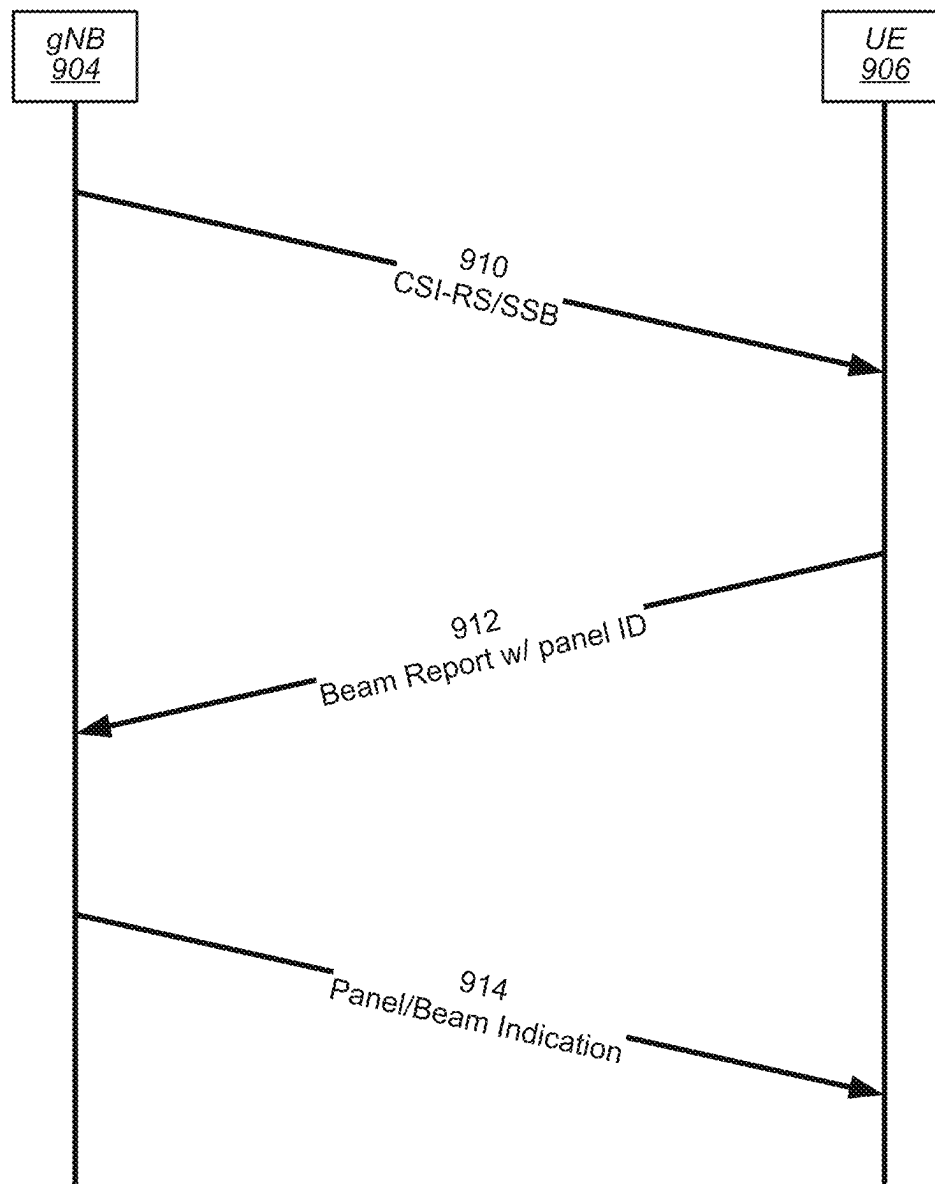


FIG. 9

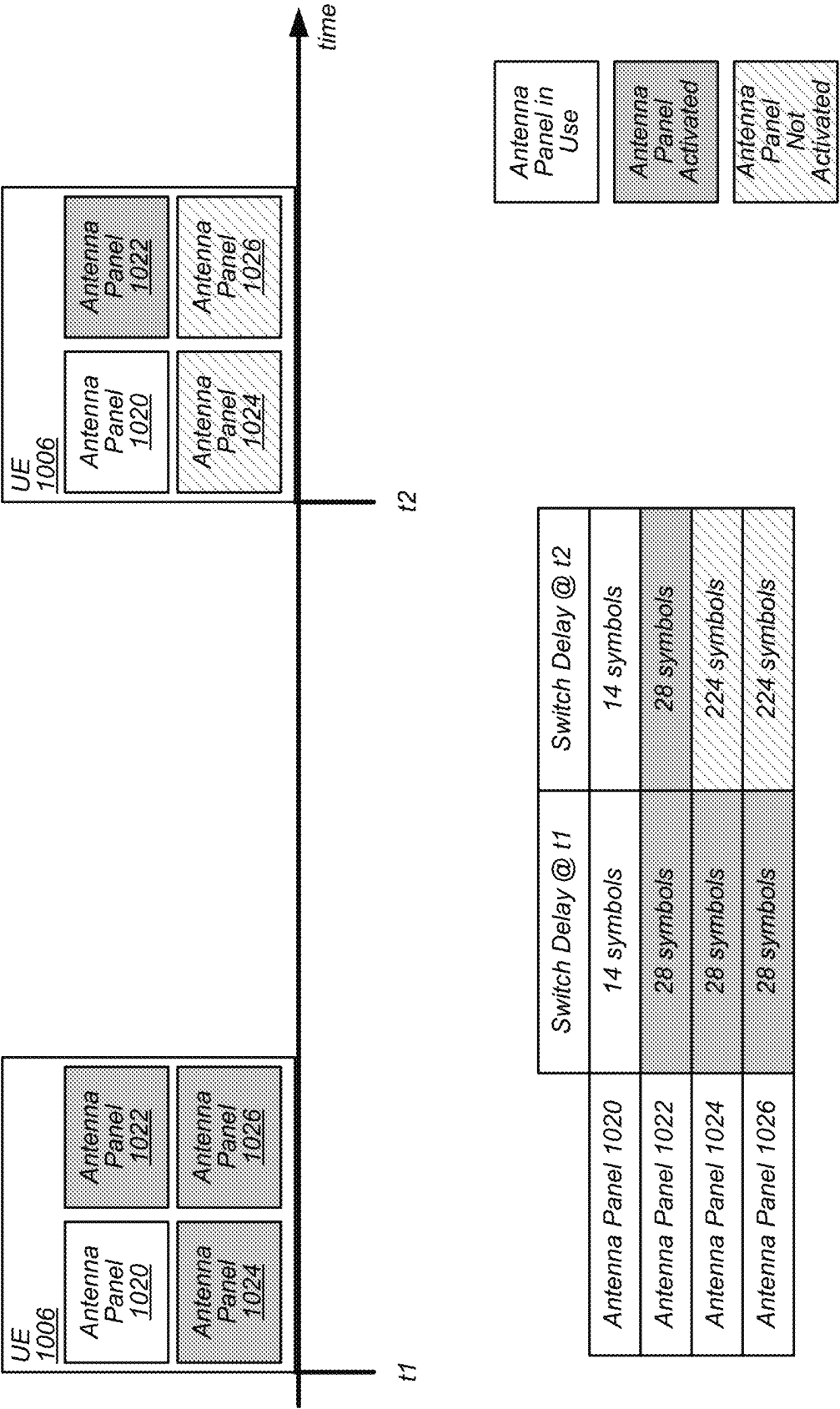


FIG. 10

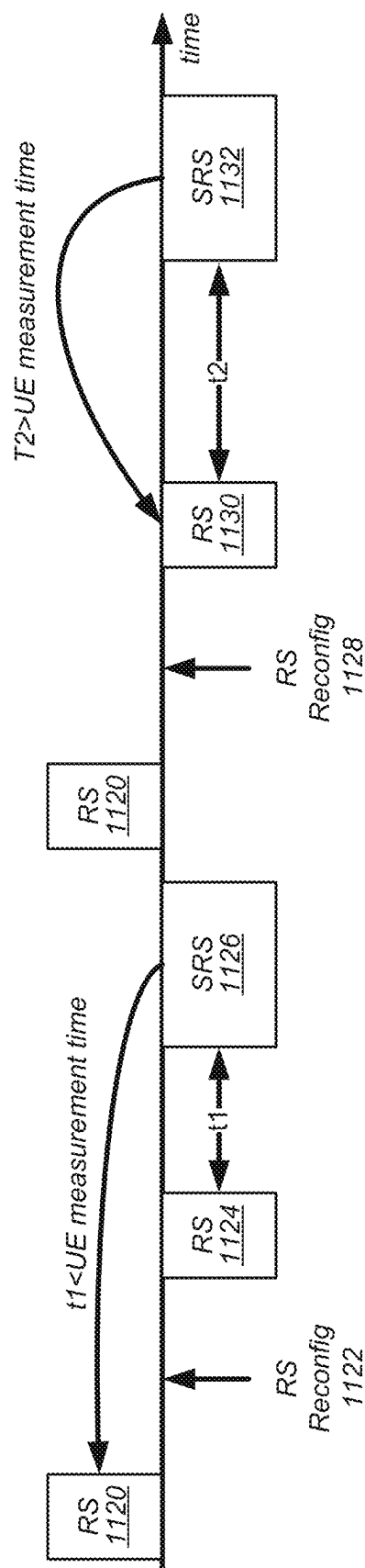


FIG. 11

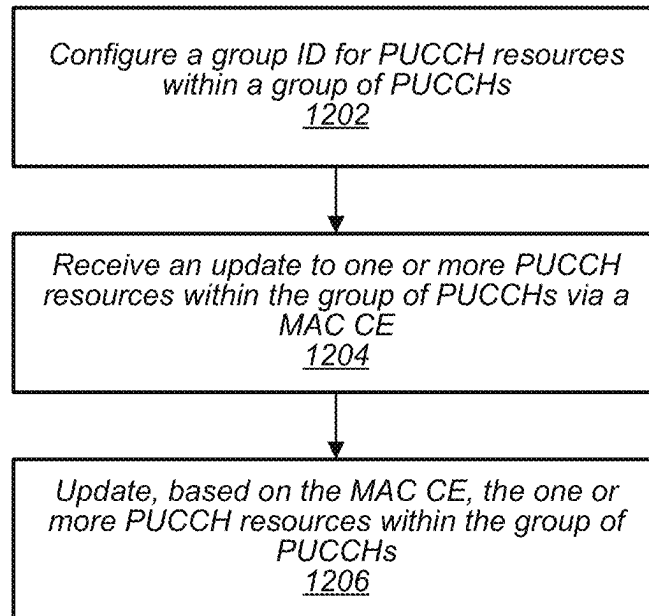


FIG. 12

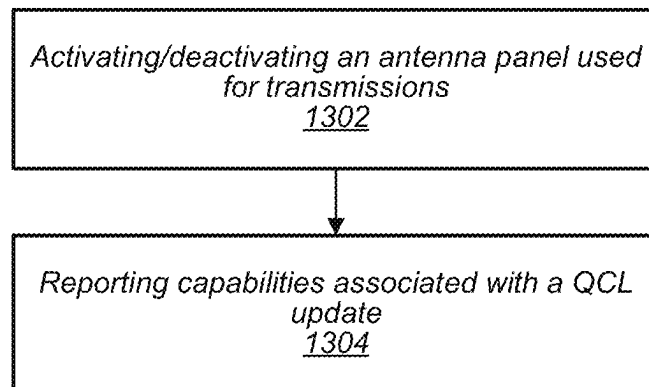


FIG. 13

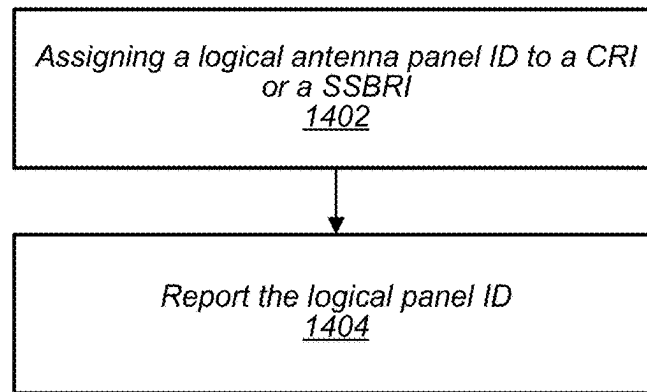


FIG. 14

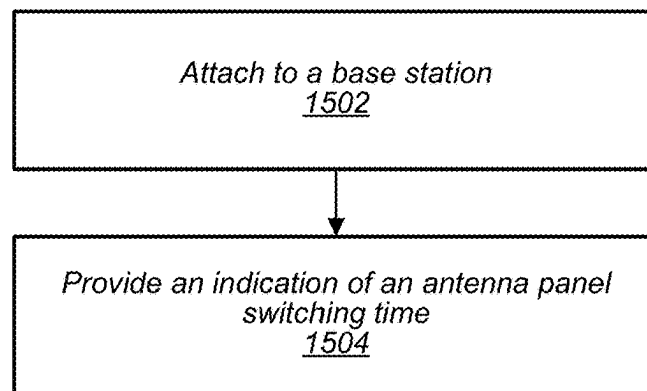


FIG. 15

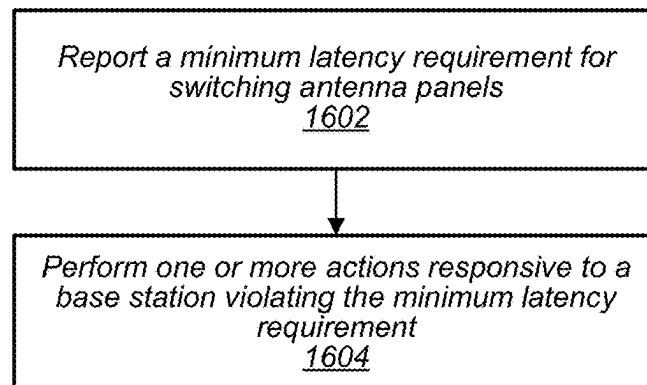


FIG. 16

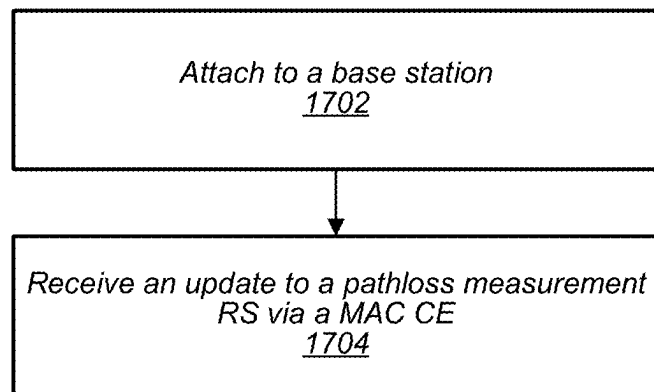


FIG. 17

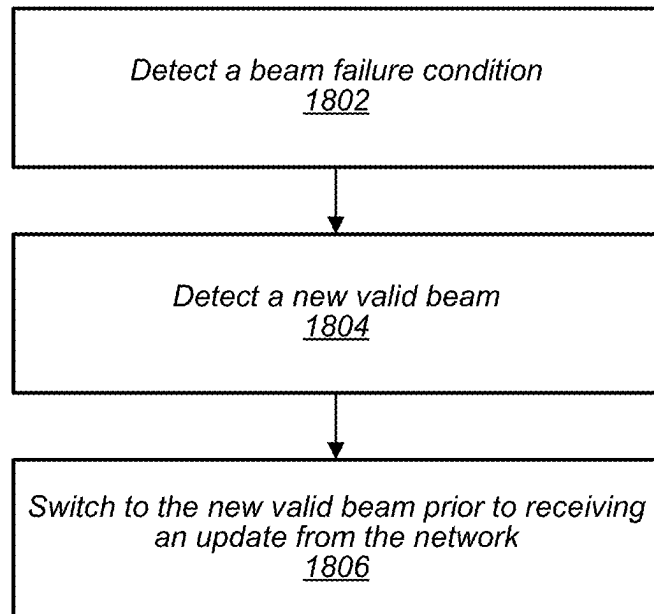


FIG. 18

5G NR FR2 BEAM MANAGEMENT ENHANCEMENTS

PRIORITY DATA

This application is a continuation of U.S. patent application Ser. No. 17/731,995, titled “5G NR FR2 Beam Management Enhancements”, filed Apr. 28, 2022, which is a continuation of U.S. patent application Ser. No. 17/021,136, titled “5G NR FR2 Beam Management Enhancements”, filed Sep. 15, 2020, which claims benefit of priority to U.S. Provisional Application Ser. No. 62/939,215, titled “5G NR FR2 Beam Management Enhancements”, filed Nov. 22, 2019, which is hereby incorporated by reference in its entirety as though fully and completely set forth herein.

The claims in the instant application are different than those of the parent application and/or other related applications. The Applicant therefore rescinds any disclaimer of claim scope made in the parent application and/or any predecessor application in relation to the instant application. Any such previous disclaimer and the cited references that it was made to avoid, may need to be revisited. Further, any disclaimer made in the instant application should not be read into or against the parent application and/or other related applications.

FIELD

The present application relates to wireless devices, and more particularly to apparatuses, systems, and methods for a wireless device to perform methods for Fifth Generation New Radio (5G NR) frequency range two (FR2) based communications beam management.

DESCRIPTION OF THE RELATED ART

Wireless communication systems are rapidly growing in usage. In recent years, wireless devices such as smart phones and tablet computers have become increasingly sophisticated. In addition to supporting telephone calls, many mobile devices now provide access to the internet, email, text messaging, and navigation using the global positioning system (GPS), and are capable of operating sophisticated applications that utilize these functionalities.

Long Term Evolution (LTE) has become the technology of choice for the majority of wireless network operators worldwide, providing mobile broadband data and high-speed Internet access to their subscriber base. LTE defines a number of downlink (DL) physical channels, categorized as transport or control channels, to carry information blocks received from medium access control (MAC) and higher layers. LTE also defines a number of physical layer channels for the uplink (UL).

For example, LTE defines a Physical Downlink Shared Channel (PDSCH) as a DL transport channel. The PDSCH is the main data-bearing channel allocated to users on a dynamic and opportunistic basis. The PDSCH carries data in Transport Blocks (TB) corresponding to a MAC protocol data unit (PDU), passed from the MAC layer to the physical (PHY) layer once per Transmission Time Interval (TTI). The PDSCH is also used to transmit broadcast information such as System Information Blocks (SIB) and paging messages.

As another example, LTE defines a Physical Downlink Control Channel (PDCCH) as a DL control channel that carries the resource assignment for UEs that are contained in a Downlink Control Information (DCI) message. Multiple PDCCHs can be transmitted in the same subframe using

Control Channel Elements (CCE), each of which is a nine set of four resource elements known as Resource Element Groups (REG). The PDCCH employs quadrature phase-shift keying (QPSK) modulation, with four QPSK symbols mapped to each REG. Furthermore, 1, 2, 4, or 8 CCEs can be used for a UE, depending on channel conditions, to ensure sufficient robustness.

Additionally, LTE defines a Physical Uplink Shared Channel (PUSCH) as a UL channel shared by all devices (user equipment, UE) in a radio cell to transmit user data to the network. The scheduling for all UEs is under control of the LTE base station (enhanced Node B, or eNB). The eNB uses the uplink scheduling grant (DCI format 0) to inform the UE about resource block (RB) assignment, and the modulation and coding scheme to be used. PUSCH typically supports QPSK and quadrature amplitude modulation (QAM). In addition to user data, the PUSCH also carries any control information necessary to decode the information, such as transport format indicators and multiple-in multiple-out (MIMO) parameters. Control data is multiplexed with information data prior to digital Fourier transform (DFT) spreading.

A proposed next telecommunications standard moving beyond the current International Mobile Telecommunications-Advanced (IMT-Advanced) Standards is called 5th generation mobile networks or 5th generation wireless systems, or 5G for short (otherwise known as 5G-NR for 5G New Radio, also simply referred to as NR). 5G-NR proposes a higher capacity for a higher density of mobile broadband users, also supporting device-to-device, ultra-reliable, and massive machine communications, as well as lower latency and lower battery consumption, than current LTE standards. Further, the 5G-NR standard may allow for less restrictive UE scheduling as compared to current LTE standards. Consequently, efforts are being made in ongoing developments of 5G-NR to take advantage of higher throughputs possible at higher frequencies.

SUMMARY

Embodiments relate to apparatuses, systems, and methods for a UE to perform Fifth Generation New Radio (5G NR) frequency range two (FR2) based communications beam management.

In some embodiments, a wireless device, e.g., such as a user equipment device (UE), may be configured to configure, via communications with a base station serving the UE, a group identifier (ID) for physical uplink control channel (PUCCH) resources within a group of PUCCHs, e.g., for FR2 based communications. In some embodiments, configuring the group ID may include comprises exchanging radio resource control (RRC) messages with the base station, e.g., during a PUCCH configuration procedure. Additionally, the UE may receive, from the base station, an update to one or more PUCCH resources within the group of PUCCHs via a medium access control (MAC) control element (CE). In some embodiments, the MAC CE may include one of a resource ID indicating an update to a PUCCH resource or the group ID to update the group of PUCCH resources. In some embodiments, the UE may update, based on the MAC-CE, the one or more PUCCH resources within the group of PUCCHs.

The techniques described herein may be implemented in and/or used with a number of different types of devices, including but not limited to cellular phones, tablet computers, wearable computing devices, portable media players, and any of various other computing devices.

This Summary is intended to provide a brief overview of some of the subject matter described in this document. Accordingly, it will be appreciated that the above-described features are merely examples and should not be construed to narrow the scope or spirit of the subject matter described herein in any way. Other features, aspects, and advantages of the subject matter described herein will become apparent from the following Detailed Description, Figures, and Claims.

BRIEF DESCRIPTION OF THE DRAWINGS

A better understanding of the present subject matter can be obtained when the following detailed description of various embodiments is considered in conjunction with the following drawings, in which:

FIG. 1A illustrates an example wireless communication system according to some embodiments.

FIG. 1B illustrates an example of a base station (BS) and an access point in communication with a user equipment (UE) device according to some embodiments.

FIG. 2 illustrates an example simplified block diagram of a WLAN Access Point (AP), according to some embodiments.

FIG. 3 illustrates an example block diagram of a UE according to some embodiments.

FIG. 4 illustrates an example block diagram of a BS according to some embodiments.

FIG. 5 illustrates an example block diagram of cellular communication circuitry, according to some embodiments.

FIG. 6A illustrates an example of connections between an EPC network, an LTE base station (eNB), and a 5G NR base station (gNB).

FIG. 6B illustrates an example of a protocol stack for an eNB and a gNB.

FIG. 7A illustrates an example of a 5G network architecture that incorporates both 3GPP (e.g., cellular) and non-3GPP (e.g., non-cellular) access to the 5G CN, according to some embodiments.

FIG. 7B illustrates an example of a 5G network architecture that incorporates both dual 3GPP (e.g., LTE and 5G NR) access and non-3GPP access to the 5G CN, according to some embodiments.

FIG. 8 illustrates an example of a baseband processor architecture for a UE, according to some embodiments.

FIG. 9 illustrates signaling between a base station and a UE to establish a beam and associated logical antenna panel ID, according to some embodiments.

FIG. 10 illustrates a multiple antenna panel UE switching panel states, according to some embodiments.

FIG. 11 illustrates a timeline for pathloss measurement RS reconfiguration, according to some embodiments.

FIGS. 12-18 illustrate block diagrams of examples of methods for Fifth Generation New Radio (5G NR) frequency range two (FR2) based communications beam management, according to some embodiments.

While the features described herein may be susceptible to various modifications and alternative forms, specific embodiments thereof are shown by way of example in the drawings and are herein described in detail. It should be understood, however, that the drawings and detailed description thereto are not intended to be limiting to the particular form disclosed, but on the contrary, the intention is to cover all modifications, equivalents and alternatives falling within the spirit and scope of the subject matter as defined by the appended claims.

DETAILED DESCRIPTION

Terms

The following is a glossary of terms used in this disclosure:

Memory Medium—Any of various types of non-transitory memory devices or storage devices. The term “memory medium” is intended to include an installation medium, e.g., a CD-ROM, floppy disks, or tape device; a computer system memory or random-access memory such as DRAM, DDR RAM, SRAM, EDO RAM, Rambus RAM, etc.; a non-volatile memory such as a Flash, magnetic media, e.g., a hard drive, or optical storage; registers, or other similar types of memory elements, etc. The memory medium may include other types of non-transitory memory as well or combinations thereof. In addition, the memory medium may be located in a first computer system in which the programs are executed, or may be located in a second different computer system which connects to the first computer system over a network, such as the Internet. In the latter instance, the second computer system may provide program instructions to the first computer for execution. The term “memory medium” may include two or more memory mediums which may reside in different locations, e.g., in different computer systems that are connected over a network. The memory medium may store program instructions (e.g., embodied as computer programs) that may be executed by one or more processors.

Carrier Medium—a memory medium as described above, as well as a physical transmission medium, such as a bus, network, and/or other physical transmission medium that conveys signals such as electrical, electromagnetic, or digital signals.

Programmable Hardware Element—includes various hardware devices comprising multiple programmable function blocks connected via a programmable interconnect. Examples include FPGAs (Field Programmable Gate Arrays), PLDs (Programmable Logic Devices), FPGAs (Field Programmable Object Arrays), and CPLDs (Complex PLDs). The programmable function blocks may range from fine grained (combinatorial logic or look up tables) to coarse grained (arithmetic logic units or processor cores). A programmable hardware element may also be referred to as “reconfigurable logic”.

Computer System—any of various types of computing or processing systems, including a personal computer system (PC), mainframe computer system, workstation, network appliance, Internet appliance, personal digital assistant (PDA), television system, grid computing system, or other device or combinations of devices. In general, the term “computer system” can be broadly defined to encompass any device (or combination of devices) having at least one processor that executes instructions from a memory medium.

User Equipment (UE) (or “UE Device”)—any of various types of computer systems devices which are mobile or portable and which performs wireless communications. Examples of UE devices include mobile telephones or smart phones (e.g., iPhone™, Android™-based phones), portable gaming devices (e.g., Nintendo DS™ PlayStation Portable™, Gameboy Advance™, iPhone™), laptops, wearable devices (e.g. smart watch, smart glasses), PDAs, portable Internet devices, music players, data storage devices, or other handheld devices, etc. In general, the term “UE” or “UE device” can be broadly defined to encompass any electronic, computing, and/or telecommunications device

(or combination of devices) which is easily transported by a user and capable of wireless communication.

Base Station—The term “Base Station” has the full breadth of its ordinary meaning, and at least includes a wireless communication station installed at a fixed location and used to communicate as part of a wireless telephone system or radio system.

Processing Element—refers to various elements or combinations of elements that are capable of performing a function in a device, such as a user equipment or a cellular network device. Processing elements may include, for example: processors and associated memory, portions or circuits of individual processor cores, entire processor cores, processor arrays, circuits such as an ASIC (Application Specific Integrated Circuit), programmable hardware elements such as a field programmable gate array (FPGA), as well as any of various combinations of the above.

Channel—a medium used to convey information from a sender (transmitter) to a receiver. It should be noted that since characteristics of the term “channel” may differ according to different wireless protocols, the term “channel” as used herein may be considered as being used in a manner that is consistent with the standard of the type of device with reference to which the term is used. In some standards, channel widths may be variable (e.g., depending on device capability, band conditions, etc.). For example, LTE may support scalable channel bandwidths from 1.4 MHz to 20 MHz. In contrast, WLAN channels may be 22 MHz wide while Bluetooth channels may be 1 Mhz wide. Other protocols and standards may include different definitions of channels. Furthermore, some standards may define and use multiple types of channels, e.g., different channels for uplink or downlink and/or different channels for different uses such as data, control information, etc.

Band—The term “band” has the full breadth of its ordinary meaning, and at least includes a section of spectrum (e.g., radio frequency spectrum) in which channels are used or set aside for the same purpose.

Automatically—refers to an action or operation performed by a computer system (e.g., software executed by the computer system) or device (e.g., circuitry, programmable hardware elements, ASICs, etc.), without user input directly specifying or performing the action or operation. Thus the term “automatically” is in contrast to an operation being manually performed or specified by the user, where the user provides input to directly perform the operation. An automatic procedure may be initiated by input provided by the user, but the subsequent actions that are performed “automatically” are not specified by the user, i.e., are not performed “manually”, where the user specifies each action to perform. For example, a user filling out an electronic form by selecting each field and providing input specifying information (e.g., by typing information, selecting check boxes, radio selections, etc.) is filling out the form manually, even though the computer system must update the form in response to the user actions. The form may be automatically filled out by the computer system where the computer system (e.g., software executing on the computer system) analyzes the fields of the form and fills in the form without any user input specifying the answers to the fields. As indicated above, the user may invoke the automatic filling of the form, but is not involved in the actual filling of the form (e.g., the user is not manually specifying answers to fields but rather they are being automatically completed). The present specification provides various examples of operations being automatically performed in response to actions the user has taken.

Approximately—refers to a value that is almost correct or exact. For example, approximately may refer to a value that is within 1 to 10 percent of the exact (or desired) value. It should be noted, however, that the actual threshold value (or tolerance) may be application dependent. For example, in some embodiments, “approximately” may mean within 0.1% of some specified or desired value, while in various other embodiments, the threshold may be, for example, 2%, 3%, 5%, and so forth, as desired or as required by the particular application.

Concurrent—refers to parallel execution or performance, where tasks, processes, or programs are performed in an at least partially overlapping manner. For example, concurrency may be implemented using “strong” or strict parallelism, where tasks are performed (at least partially) in parallel on respective computational elements, or using “weak parallelism”, where the tasks are performed in an interleaved manner, e.g., by time multiplexing of execution threads.

Various components may be described as “configured to” perform a task or tasks. In such contexts, “configured to” is a broad recitation generally meaning “having structure that” performs the task or tasks during operation. As such, the component can be configured to perform the task even when the component is not currently performing that task (e.g., a set of electrical conductors may be configured to electrically connect a module to another module, even when the two modules are not connected). In some contexts, “configured to” may be a broad recitation of structure generally meaning “having circuitry that” performs the task or tasks during operation. As such, the component can be configured to perform the task even when the component is not currently on. In general, the circuitry that forms the structure corresponding to “configured to” may include hardware circuits.

Various components may be described as performing a task or tasks, for convenience in the description. Such descriptions should be interpreted as including the phrase “configured to.” Reciting a component that is configured to perform one or more tasks is expressly intended not to invoke 35 U.S.C. § 112(f) interpretation for that component. FIGS. 1A and 1B—Communication Systems

FIG. 1A illustrates a simplified example wireless communication system, according to some embodiments. It is noted that the system of FIG. 1 is merely one example of a possible system, and that features of this disclosure may be implemented in any of various systems, as desired.

As shown, the example wireless communication system includes a base station **102A** which communicates over a transmission medium with one or more user devices **106A**, **106B**, etc., through **106N**. Each of the user devices may be referred to herein as a “user equipment” (UE). Thus, the user devices **106** are referred to as UEs or UE devices.

The base station (BS) **102A** may be a base transceiver station (BTS) or cell site (a “cellular base station”) and may include hardware that enables wireless communication with the UEs **106A** through **106N**.

The communication area (or coverage area) of the base station may be referred to as a “cell.” The base station **102A** and the UEs **106** may be configured to communicate over the transmission medium using any of various radio access technologies (RATs), also referred to as wireless communication technologies, or telecommunication standards, such as GSM, UMTS (associated with, for example, WCDMA or TD-SCDMA air interfaces), LTE, LTE-Advanced (LTE-A), 5G new radio (5G NR), HSPA, 3GPP2 CDMA2000 (e.g., 1×RTT, 1×EV-DO, HRPD, eHRPD), etc. Note that if the base station **102A** is implemented in the context of LTE, it may alternately be referred to as an ‘eNodeB’ or ‘eNB’. Note

that if the base station **102A** is implemented in the context of 5G NR, it may alternately be referred to as ‘gNodeB’ or ‘gNB’.

As shown, the base station **102A** may also be equipped to communicate with a network **100** (e.g., a core network of a cellular service provider, a telecommunication network such as a public switched telephone network (PSTN), and/or the Internet, among various possibilities). Thus, the base station **102A** may facilitate communication between the user devices and/or between the user devices and the network **100**. In particular, the cellular base station **102A** may provide UEs **106** with various telecommunication capabilities, such as voice, SMS and/or data services.

Base station **102A** and other similar base stations (such as base stations **102B** . . . **102N**) operating according to the same or a different cellular communication standard may thus be provided as a network of cells, which may provide continuous or nearly continuous overlapping service to UEs **106A-N** and similar devices over a geographic area via one or more cellular communication standards.

Thus, while base station **102A** may act as a “serving cell” for UEs **106A-N** as illustrated in FIG. 1, each UE **106** may also be capable of receiving signals from (and possibly within communication range of) one or more other cells (which might be provided by base stations **102B-N** and/or any other base stations), which may be referred to as “neighboring cells”. Such cells may also be capable of facilitating communication between user devices and/or between user devices and the network **100**. Such cells may include “macro” cells, “micro” cells, “pico” cells, and/or cells which provide any of various other granularities of service area size. For example, base stations **102A-B** illustrated in FIG. 1 might be macro cells, while base station **102N** might be a micro cell. Other configurations are also possible.

In some embodiments, base station **102A** may be a next generation base station, e.g., a 5G New Radio (5G NR) base station, or “gNB”. In some embodiments, a gNB may be connected to a legacy evolved packet core (EPC) network and/or to a NR core (NRC) network. In addition, a gNB cell may include one or more transition and reception points (TRPs). In addition, a UE capable of operating according to 5G NR may be connected to one or more TRPs within one or more gNBs.

Note that a UE **106** may be capable of communicating using multiple wireless communication standards. For example, the UE **106** may be configured to communicate using a wireless networking (e.g., Wi-Fi) and/or peer-to-peer wireless communication protocol (e.g., Bluetooth, Wi-Fi peer-to-peer, etc.) in addition to at least one cellular communication protocol (e.g., GSM, UMTS (associated with, for example, WCDMA or TD-SCDMA air interfaces), LTE, LTE-A, 5G NR, HSPA, 3GPP2 CDMA2000 (e.g., 1×RTT, 1×EV-DO, HRPD, eHRPD), etc.). The UE **106** may also or alternatively be configured to communicate using one or more global navigational satellite systems (GNSS, e.g., GPS or GLONASS), one or more mobile television broadcasting standards (e.g., ATSC-M/H or DVB-H), and/or any other wireless communication protocol, if desired. Other combinations of wireless communication standards (including more than two wireless communication standards) are also possible.

FIG. 1B illustrates user equipment **106** (e.g., one of the devices **106A** through **106N**) in communication with a base station **102** and an access point **112**, according to some embodiments. The UE **106** may be a device with both cellular communication capability and non-cellular commu-

nication capability (e.g., Bluetooth, Wi-Fi, and so forth) such as a mobile phone, a hand-held device, a computer or a tablet, or virtually any type of wireless device.

The UE **106** may include a processor that is configured to execute program instructions stored in memory. The UE **106** may perform any of the method embodiments described herein by executing such stored instructions. Alternatively, or in addition, the UE **106** may include a programmable hardware element such as an FPGA (field-programmable gate array) that is configured to perform any of the method embodiments described herein, or any portion of any of the method embodiments described herein.

The UE **106** may include one or more antennas for communicating using one or more wireless communication protocols or technologies. In some embodiments, the UE **106** may be configured to communicate using, for example, CDMA2000 (1×RTT/1×EV-DO/HRPD/eHRPD), LTE/LTE-Advanced, or 5G NR using a single shared radio and/or GSM, LTE, LTE-Advanced, or 5G NR using the single shared radio. The shared radio may couple to a single antenna, or may couple to multiple antennas (e.g., for MIMO) for performing wireless communications. In general, a radio may include any combination of a baseband processor, analog RF signal processing circuitry (e.g., including filters, mixers, oscillators, amplifiers, etc.), or digital processing circuitry (e.g., for digital modulation as well as other digital processing). Similarly, the radio may implement one or more receive and transmit chains using the aforementioned hardware. For example, the UE **106** may share one or more parts of a receive and/or transmit chain between multiple wireless communication technologies, such as those discussed above.

In some embodiments, the UE **106** may include separate transmit and/or receive chains (e.g., including separate antennas and other radio components) for each wireless communication protocol with which it is configured to communicate. As a further possibility, the UE **106** may include one or more radios which are shared between multiple wireless communication protocols, and one or more radios which are used exclusively by a single wireless communication protocol. For example, the UE **106** might include a shared radio for communicating using either of LTE or 5G NR (or LTE or 1×RTT or LTE or GSM), and separate radios for communicating using each of Wi-Fi and Bluetooth. Other configurations are also possible.

FIG. 2—Access Point Block Diagram

FIG. 2 illustrates an exemplary block diagram of an access point (AP) **112**. It is noted that the block diagram of the AP of FIG. 2 is only one example of a possible system. As shown, the AP **112** may include processor(s) **204** which may execute program instructions for the AP **112**. The processor(s) **204** may also be coupled (directly or indirectly) to memory management unit (MMU) **240**, which may be configured to receive addresses from the processor(s) **204** and to translate those addresses to locations in memory (e.g., memory **260** and read only memory (ROM) **250**) or to other circuits or devices.

The AP **112** may include at least one network port **270**. The network port **270** may be configured to couple to a wired network and provide a plurality of devices, such as UEs **106**, access to the Internet. For example, the network port **270** (or an additional network port) may be configured to couple to a local network, such as a home network or an enterprise network. For example, port **270** may be an Ethernet port. The local network may provide connectivity to additional networks, such as the Internet.

The AP **112** may include at least one antenna **234**, which may be configured to operate as a wireless transceiver and may be further configured to communicate with UE **106** via wireless communication circuitry **230**. The antenna **234** communicates with the wireless communication circuitry **230** via communication chain **232**. Communication chain **232** may include one or more receive chains, one or more transmit chains or both. The wireless communication circuitry **230** may be configured to communicate via Wi-Fi or WLAN, e.g., 802.11. The wireless communication circuitry **230** may also, or alternatively, be configured to communicate via various other wireless communication technologies, including, but not limited to, 5G NR, Long-Term Evolution (LTE), LTE Advanced (LTE-A), Global System for Mobile (GSM), Wideband Code Division Multiple Access (WCDMA), CDMA2000, etc., for example when the AP is co-located with a base station in case of a small cell, or in other instances when it may be desirable for the AP **112** to communicate via various different wireless communication technologies.

In some embodiments, as further described below, an AP **112** may be configured to perform methods for 5G NR FR2 based communications beam management as further described herein.

FIG. 3—Block Diagram of a UE

FIG. 3 illustrates an example simplified block diagram of a communication device **106**, according to some embodiments. It is noted that the block diagram of the communication device of FIG. 3 is only one example of a possible communication device. According to embodiments, communication device **106** may be a user equipment (UE) device, a mobile device or mobile station, a wireless device or wireless station, a desktop computer or computing device, a mobile computing device (e.g., a laptop, notebook, or portable computing device), a tablet and/or a combination of devices, among other devices. As shown, the communication device **106** may include a set of components **300** configured to perform core functions. For example, this set of components may be implemented as a system on chip (SOC), which may include portions for various purposes. Alternatively, this set of components **300** may be implemented as separate components or groups of components for the various purposes. The set of components **300** may be coupled (e.g., communicatively; directly or indirectly) to various other circuits of the communication device **106**.

For example, the communication device **106** may include various types of memory (e.g., including NAND flash **310**), an input/output interface such as connector **320** (e.g., for connecting to a computer system; dock; charging station; input devices, such as a microphone, camera, keyboard; output devices, such as speakers; etc.), the display **360**, which may be integrated with or external to the communication device **106**, and cellular communication circuitry **330** such as for 5G NR, LTE, GSM, etc., and short to medium range wireless communication circuitry **329** (e.g., Bluetooth™ and WLAN circuitry). In some embodiments, communication device **106** may include wired communication circuitry (not shown), such as a network interface card, e.g., for Ethernet.

The cellular communication circuitry **330** may couple (e.g., communicatively; directly or indirectly) to one or more antennas, such as antennas **335** and **336** as shown. The short to medium range wireless communication circuitry **329** may also couple (e.g., communicatively; directly or indirectly) to one or more antennas, such as antennas **337** and **338** as shown. Alternatively, the short to medium range wireless communication circuitry **329** may couple (e.g.,

communicatively; directly or indirectly) to the antennas **335** and **336** in addition to, or instead of, coupling (e.g., communicatively; directly or indirectly) to the antennas **337** and **338**. The short to medium range wireless communication circuitry **329** and/or cellular communication circuitry **330** may include multiple receive chains and/or multiple transmit chains for receiving and/or transmitting multiple spatial streams, such as in a multiple-input multiple output (MIMO) configuration.

In some embodiments, as further described below, cellular communication circuitry **330** may include dedicated receive chains (including and/or coupled to, e.g., communicatively; directly or indirectly, dedicated processors and/or radios) for multiple RATs (e.g., a first receive chain for LTE and a second receive chain for 5G NR). In addition, in some embodiments, cellular communication circuitry **330** may include a single transmit chain that may be switched between radios dedicated to specific RATs. For example, a first radio may be dedicated to a first RAT, e.g., LTE, and may be in communication with a dedicated receive chain and a transmit chain shared with an additional radio, e.g., a second radio that may be dedicated to a second RAT, e.g., 5G NR, and may be in communication with a dedicated receive chain and the shared transmit chain.

The communication device **106** may also include and/or be configured for use with one or more user interface elements. The user interface elements may include any of various elements, such as display **360** (which may be a touchscreen display), a keyboard (which may be a discrete keyboard or may be implemented as part of a touchscreen display), a mouse, a microphone and/or speakers, one or more cameras, one or more buttons, and/or any of various other elements capable of providing information to a user and/or receiving or interpreting user input.

The communication device **106** may further include one or more smart cards **345** that include SIM (Subscriber Identity Module) functionality, such as one or more UICC(s) (Universal Integrated Circuit Card(s)) cards **345**.

As shown, the SOC **300** may include processor(s) **302**, which may execute program instructions for the communication device **106** and display circuitry **304**, which may perform graphics processing and provide display signals to the display **360**. The processor(s) **302** may also be coupled to memory management unit (MMU) **340**, which may be configured to receive addresses from the processor(s) **302** and translate those addresses to locations in memory (e.g., memory **306**, read only memory (ROM) **350**, NAND flash memory **310**) and/or to other circuits or devices, such as the display circuitry **304**, short to medium range wireless communication circuitry **329**, cellular communication circuitry **330**, connector I/F **320**, and/or display **360**. The MMU **340** may be configured to perform memory protection and page table translation or set up. In some embodiments, the MMU **340** may be included as a portion of the processor(s) **302**.

As noted above, the communication device **106** may be configured to communicate using wireless and/or wired communication circuitry. The communication device **106** may be configured to perform methods for 5G NR FR2 based communications beam management as further described herein.

As described herein, the communication device **106** may include hardware and software components for implementing the above features for a communication device **106** to communicate a scheduling profile for power savings to a network. The processor **302** of the communication device **106** may be configured to implement part or all of the features described herein, e.g., by executing program

instructions stored on a memory medium (e.g., a non-transitory computer-readable memory medium). Alternatively (or in addition), processor **302** may be configured as a programmable hardware element, such as an FPGA (Field Programmable Gate Array), or as an ASIC (Application Specific Integrated Circuit). Alternatively (or in addition) the processor **302** of the communication device **106**, in conjunction with one or more of the other components **300**, **304**, **306**, **310**, **320**, **329**, **330**, **340**, **345**, **350**, **360** may be configured to implement part or all of the features described herein.

In addition, as described herein, processor **302** may include one or more processing elements. Thus, processor **302** may include one or more integrated circuits (ICs) that are configured to perform the functions of processor **302**. In addition, each integrated circuit may include circuitry (e.g., first circuitry, second circuitry, etc.) configured to perform the functions of processor(s) **302**.

Further, as described herein, cellular communication circuitry **330** and short to medium range wireless communication circuitry **329** may each include one or more processing elements. In other words, one or more processing elements may be included in cellular communication circuitry **330** and, similarly, one or more processing elements may be included in short to medium range wireless communication circuitry **329**. Thus, cellular communication circuitry **330** may include one or more integrated circuits (ICs) that are configured to perform the functions of cellular communication circuitry **330**. In addition, each integrated circuit may include circuitry (e.g., first circuitry, second circuitry, etc.) configured to perform the functions of cellular communication circuitry **330**. Similarly, the short to medium range wireless communication circuitry **329** may include one or more ICs that are configured to perform the functions of short to medium range wireless communication circuitry **329**. In addition, each integrated circuit may include circuitry (e.g., first circuitry, second circuitry, etc.) configured to perform the functions of short to medium range wireless communication circuitry **329**.

FIG. 4—Block Diagram of a Base Station

FIG. 4 illustrates an example block diagram of a base station **102**, according to some embodiments. It is noted that the base station of FIG. 4 is merely one example of a possible base station. As shown, the base station **102** may include processor(s) **404** which may execute program instructions for the base station **102**. The processor(s) **404** may also be coupled to memory management unit (MMU) **440**, which may be configured to receive addresses from the processor(s) **404** and translate those addresses to locations in memory (e.g., memory **460** and read only memory (ROM) **450**) or to other circuits or devices.

The base station **102** may include at least one network port **470**. The network port **470** may be configured to couple to a telephone network and provide a plurality of devices, such as UE devices **106**, access to the telephone network as described above in FIGS. 1 and 2.

The network port **470** (or an additional network port) may also or alternatively be configured to couple to a cellular network, e.g., a core network of a cellular service provider. The core network may provide mobility related services and/or other services to a plurality of devices, such as UE devices **106**. In some cases, the network port **470** may couple to a telephone network via the core network, and/or the core network may provide a telephone network (e.g., among other UE devices serviced by the cellular service provider).

In some embodiments, base station **102** may be a next generation base station, e.g., a 5G New Radio (5G NR) base station, or “gNB”. In such embodiments, base station **102** may be connected to a legacy evolved packet core (EPC) network and/or to a NR core (NRC) network. In addition, base station **102** may be considered a 5G NR cell and may include one or more transition and reception points (TRPs). In addition, a UE capable of operating according to 5G NR may be connected to one or more TRPs within one or more gNBs.

The base station **102** may include at least one antenna **434**, and possibly multiple antennas. The at least one antenna **434** may be configured to operate as a wireless transceiver and may be further configured to communicate with UE devices **106** via radio **430**. The antenna **434** communicates with the radio **430** via communication chain **432**. Communication chain **432** may be a receive chain, a transmit chain or both. The radio **430** may be configured to communicate via various wireless communication standards, including, but not limited to, 5G NR, LTE, LTE-A, GSM, UMTS, CDMA2000, Wi-Fi, etc.

The base station **102** may be configured to communicate wirelessly using multiple wireless communication standards. In some instances, the base station **102** may include multiple radios, which may enable the base station **102** to communicate according to multiple wireless communication technologies. For example, as one possibility, the base station **102** may include an LTE radio for performing communication according to LTE as well as a 5G NR radio for performing communication according to 5G NR. In such a case, the base station **102** may be capable of operating as both an LTE base station and a 5G NR base station. As another possibility, the base station **102** may include a multi-mode radio which is capable of performing communications according to any of multiple wireless communication technologies (e.g., 5G NR and Wi-Fi, LTE and Wi-Fi, LTE and UMTS, LTE and CDMA2000, UMTS and GSM, etc.).

As described further subsequently herein, the BS **102** may include hardware and software components for implementing or supporting implementation of features described herein. The processor **404** of the base station **102** may be configured to implement or support implementation of part or all of the methods described herein, e.g., by executing program instructions stored on a memory medium (e.g., a non-transitory computer-readable memory medium). Alternatively, the processor **404** may be configured as a programmable hardware element, such as an FPGA (Field Programmable Gate Array), or as an ASIC (Application Specific Integrated Circuit), or a combination thereof. Alternatively (or in addition) the processor **404** of the BS **102**, in conjunction with one or more of the other components **430**, **432**, **434**, **440**, **450**, **460**, **470** may be configured to implement or support implementation of part or all of the features described herein.

In addition, as described herein, processor(s) **404** may be comprised of one or more processing elements. In other words, one or more processing elements may be included in processor(s) **404**. Thus, processor(s) **404** may include one or more integrated circuits (ICs) that are configured to perform the functions of processor(s) **404**. In addition, each integrated circuit may include circuitry (e.g., first circuitry, second circuitry, etc.) configured to perform the functions of processor(s) **404**.

Further, as described herein, radio **430** may be comprised of one or more processing elements. In other words, one or more processing elements may be included in radio **430**.

Thus, radio **430** may include one or more integrated circuits (ICs) that are configured to perform the functions of radio **430**. In addition, each integrated circuit may include circuitry (e.g., first circuitry, second circuitry, etc.) configured to perform the functions of radio **430**.

FIG. 5: Block Diagram of Cellular Communication Circuitry

FIG. 5 illustrates an example simplified block diagram of cellular communication circuitry, according to some embodiments. It is noted that the block diagram of the cellular communication circuitry of FIG. 5 is only one example of a possible cellular communication circuit. According to embodiments, cellular communication circuitry **330** may be included in a communication device, such as communication device **106** described above. As noted above, communication device **106** may be a user equipment (UE) device, a mobile device or mobile station, a wireless device or wireless station, a desktop computer or computing device, a mobile computing device (e.g., a laptop, notebook, or portable computing device), a tablet and/or a combination of devices, among other devices.

The cellular communication circuitry **330** may couple (e.g., communicatively; directly or indirectly) to one or more antennas, such as antennas **335a-b** and **336** as shown (in FIG. 3). In some embodiments, cellular communication circuitry **330** may include dedicated receive chains (including and/or coupled to, e.g., communicatively; directly or indirectly, dedicated processors and/or radios) for multiple RATs (e.g., a first receive chain for LTE and a second receive chain for 5G NR). For example, as shown in FIG. 5, cellular communication circuitry **330** may include a modem **510** and a modem **520**. Modem **510** may be configured for communications according to a first RAT, e.g., such as LTE or LTE-A, and modem **520** may be configured for communications according to a second RAT, e.g., such as 5G NR.

As shown, modem **510** may include one or more processors **512** and a memory **516** in communication with processors **512**. Modem **510** may be in communication with a radio frequency (RF) front end **530**. RF front end **530** may include circuitry for transmitting and receiving radio signals. For example, RF front end **530** may include receive circuitry (RX) **532** and transmit circuitry (TX) **534**. In some embodiments, receive circuitry **532** may be in communication with downlink (DL) front end **550**, which may include circuitry for receiving radio signals via antenna **335a**.

Similarly, modem **520** may include one or more processors **522** and a memory **526** in communication with processors **522**. Modem **520** may be in communication with an RF front end **540**. RF front end **540** may include circuitry for transmitting and receiving radio signals. For example, RF front end **540** may include receive circuitry **542** and transmit circuitry **544**. In some embodiments, receive circuitry **542** may be in communication with DL front end **560**, which may include circuitry for receiving radio signals via antenna **335b**.

In some embodiments, a switch **570** may couple transmit circuitry **534** to uplink (UL) front end **572**. In addition, switch **570** may couple transmit circuitry **544** to UL front end **572**. UL front end **572** may include circuitry for transmitting radio signals via antenna **336**. Thus, when cellular communication circuitry **330** receives instructions to transmit according to the first RAT (e.g., as supported via modem **510**), switch **570** may be switched to a first state that allows modem **510** to transmit signals according to the first RAT (e.g., via a transmit chain that includes transmit circuitry **534** and UL front end **572**). Similarly, when cellular communication circuitry **330** receives instructions to transmit accord-

ing to the second RAT (e.g., as supported via modem **520**), switch **570** may be switched to a second state that allows modem **520** to transmit signals according to the second RAT (e.g., via a transmit chain that includes transmit circuitry **544** and UL front end **572**).

In some embodiments, the cellular communication circuitry **330** may be configured to perform methods for 5G NR FR2 based communications beam management as further described herein.

As described herein, the modem **510** may include hardware and software components for implementing the above features or for time division multiplexing UL data for NSA NR operations, as well as the various other techniques described herein. The processors **512** may be configured to implement part or all of the features described herein, e.g., by executing program instructions stored on a memory medium (e.g., a non-transitory computer-readable memory medium). Alternatively (or in addition), processor **512** may be configured as a programmable hardware element, such as an FPGA (Field Programmable Gate Array), or as an ASIC (Application Specific Integrated Circuit). Alternatively (or in addition) the processor **512**, in conjunction with one or more of the other components **530**, **532**, **534**, **550**, **570**, **572**, **335** and **336** may be configured to implement part or all of the features described herein.

In addition, as described herein, processors **512** may include one or more processing elements. Thus, processors **512** may include one or more integrated circuits (ICs) that are configured to perform the functions of processors **512**. In addition, each integrated circuit may include circuitry (e.g., first circuitry, second circuitry, etc.) configured to perform the functions of processors **512**.

As described herein, the modem **520** may include hardware and software components for implementing the above features for communicating a scheduling profile for power savings to a network, as well as the various other techniques described herein. The processors **522** may be configured to implement part or all of the features described herein, e.g., by executing program instructions stored on a memory medium (e.g., a non-transitory computer-readable memory medium). Alternatively (or in addition), processor **522** may be configured as a programmable hardware element, such as an FPGA (Field Programmable Gate Array), or as an ASIC (Application Specific Integrated Circuit). Alternatively (or in addition) the processor **522**, in conjunction with one or more of the other components **540**, **542**, **544**, **550**, **570**, **572**, **335** and **336** may be configured to implement part or all of the features described herein.

In addition, as described herein, processors **522** may include one or more processing elements. Thus, processors **522** may include one or more integrated circuits (ICs) that are configured to perform the functions of processors **522**. In addition, each integrated circuit may include circuitry (e.g., first circuitry, second circuitry, etc.) configured to perform the functions of processors **522**.

5G NR Architecture with LTE

In some implementations, fifth generation (5G) wireless communication will initially be deployed concurrently with current wireless communication standards (e.g., LTE). For example, dual connectivity between LTE and 5G new radio (5G NR or NR) has been specified as part of the initial deployment of NR. Thus, as illustrated in FIGS. 6A-B, evolved packet core (EPC) network **600** may continue to communicate with current LTE base stations (e.g., eNB **602**). In addition, eNB **602** may be in communication with a 5G NR base station (e.g., gNB **604**) and may pass data between the EPC network **600** and gNB **604**. Thus, EPC

15

network 600 may be used (or reused) and gNB 604 may serve as extra capacity for UEs, e.g., for providing increased downlink throughput to UEs. In other words, LTE may be used for control plane signaling and NR may be used for user plane signaling. Thus, LTE may be used to establish connections to the network and NR may be used for data services.

FIG. 6B illustrates a proposed protocol stack for eNB 602 and gNB 604. As shown, eNB 602 may include a medium access control (MAC) layer 632 that interfaces with radio link control (RLC) layers 622a-b. RLC layer 622a may also interface with packet data convergence protocol (PDCP) layer 612a and RLC layer 622b may interface with PDCP layer 612b. Similar to dual connectivity as specified in LTE-Advanced Release 12, PDCP layer 612a may interface via a master cell group (MCG) bearer with EPC network 600 whereas PDCP layer 612b may interface via a split bearer with EPC network 600.

Additionally, as shown, gNB 604 may include a MAC layer 634 that interfaces with RLC layers 624a-b. RLC layer 624a may interface with PDCP layer 612b of eNB 602 via an X₂ interface for information exchange and/or coordination (e.g., scheduling of a UE) between eNB 602 and gNB 604. In addition, RLC layer 624b may interface with PDCP layer 614. Similar to dual connectivity as specified in LTE-Advanced Release 12, PDCP layer 614 may interface with EPC network 600 via a secondary cell group (SCG) bearer. Thus, eNB 602 may be considered a master node (MeNB) while gNB 604 may be considered a secondary node (SgNB). In some scenarios, a UE may be required to maintain a connection to both an MeNB and a SgNB. In such scenarios, the MeNB may be used to maintain a radio resource control (RRC) connection to an EPC while the SgNB may be used for capacity (e.g., additional downlink and/or uplink throughput).

5G Core Network Architecture—Interworking with Wi-Fi

In some embodiments, the 5G core network (CN) may be accessed via (or through) a cellular connection/interface (e.g., via a 3GPP communication architecture/protocol) and a non-cellular connection/interface (e.g., a non-3GPP access architecture/protocol such as Wi-Fi connection). FIG. 7A illustrates an example of a 5G network architecture that incorporates both 3GPP (e.g., cellular) and non-3GPP (e.g., non-cellular) access to the 5G CN, according to some embodiments. As shown, a user equipment device (e.g., such as UE 106) may access the 5G CN through both a radio access network (RAN, e.g., such as gNB or base station 604) and an access point, such as AP 112. The AP 112 may include a connection to the Internet 700 as well as a connection to a non-3GPP inter-working function (N3IWF) 702 network entity. The N3IWF may include a connection to a core access and mobility management function (AMF) 704 of the 5G CN. The AMF 704 may include an instance of a 5G mobility management (5G MM) function associated with the UE 106. In addition, the RAN (e.g., gNB 604) may also have a connection to the AMF 704. Thus, the 5G CN may support unified authentication over both connections as well as allow simultaneous registration for UE 106 access via both gNB 604 and AP 112. As shown, the AMF 704 may include one or more functional entities associated with the 5G CN (e.g., network slice selection function (NSSF) 720, short message service function (SMSF) 722, application function (AF) 724, unified data management (UDM) 726, policy control function (PCF) 728, and/or authentication server function (AUSF) 730). Note that these functional entities may also be supported by a session management function (SMF) 706a and an SMF 706b of the 5G CN. The

16

AMF 706 may be connected to (or in communication with) the SMF 706a. Further, the gNB 604 may in communication with (or connected to) a user plane function (UPF) 708a that may also be communication with the SMF 706a. Similarly, the N3IWF 702 may be communicating with a UPF 708b that may also be communicating with the SMF 706b. Both UPFs may be communicating with the data network (e.g., DN 710a and 710b) and/or the Internet 700 and IMS core network 710.

FIG. 7B illustrates an example of a 5G network architecture that incorporates both dual 3GPP (e.g., LTE and 5G NR) access and non-3GPP access to the 5G CN, according to some embodiments. As shown, a user equipment device (e.g., such as UE 106) may access the 5G CN through both a radio access network (RAN, e.g., such as gNB or base station 604 or eNB or base station 602) and an access point, such as AP 112. The AP 112 may include a connection to the Internet 700 as well as a connection to the N3IWF 702 network entity. The N3IWF may include a connection to the AMF 704 of the 5G CN. The AMF 704 may include an instance of the 5G MM function associated with the UE 106. In addition, the RAN (e.g., gNB 604) may also have a connection to the AMF 704. Thus, the 5G CN may support unified authentication over both connections as well as allow simultaneous registration for UE 106 access via both gNB 604 and AP 112. In addition, the 5G CN may support dual-registration of the UE on both a legacy network (e.g., LTE via base station 602) and a 5G network (e.g., via base station 604). As shown, the base station 602 may have connections to a mobility management entity (MME) 742 and a serving gateway (SGW) 744. The MME 742 may have connections to both the SGW 744 and the AMF 704. In addition, the SGW 744 may have connections to both the SMF 706a and the UPF 708a. As shown, the AMF 704 may include one or more functional entities associated with the 5G CN (e.g., NSSF 720, SMSF 722, AF 724, UDM 726, PCF 728, and/or AUSF 730). Note that UDM 726 may also include a home subscriber server (HSS) function and the PCF may also include a policy and charging rules function (PCRF). Note further that these functional entities may also be supported by the SMF 706a and the SMF 706b of the 5G CN. The AMF 706 may be connected to (or in communication with) the SMF 706a. Further, the gNB 604 may in communication with (or connected to) the UPF 708a that may also be communication with the SMF 706a. Similarly, the N3IWF 702 may be communicating with a UPF 708b that may also be communicating with the SMF 706b. Both UPFs may be communicating with the data network (e.g., DN 710a and 710b) and/or the Internet 700 and IMS core network 710.

Note that in various embodiments, one or more of the above described network entities may be configured to perform methods for 5G NR FR2 based communications beam management, e.g., as further described herein.

FIG. 8 illustrates an example of a baseband processor architecture for a UE (e.g., such as UE 106), according to some embodiments. The baseband processor architecture 800 described in FIG. 8 may be implemented on one or more radios (e.g., radios 329 and/or 330 described above) or modems (e.g., modems 510 and/or 520) as described above. As shown, the non-access stratum (NAS) 810 may include a 5G NAS 820 and a legacy NAS 850. The legacy NAS 850 may include a communication connection with a legacy access stratum (AS) 870. The 5G NAS 820 may include communication connections with both a 5G AS 840 and a non-3GPP AS 830 and Wi-Fi AS 832. The 5G NAS 820 may include functional entities associated with both access stratum

tums. Thus, the 5G NAS **820** may include multiple 5G MM entities **826** and **828** and 5G session management (SM) entities **822** and **824**. The legacy NAS **850** may include functional entities such as short message service (SMS) entity **852**, evolved packet system (EPS) session management (ESM) entity **854**, session management (SM) entity **856**, EPS mobility management (EMM) entity **858**, and mobility management (MM)/GPRS mobility management (GMM) entity **860**. In addition, the legacy AS **870** may include functional entities such as LTE AS **872**, UMTS AS **874**, and/or GSM/GPRS AS **876**.

Thus, the baseband processor architecture **800** allows for a common 5G-NAS for both 5G cellular and non-cellular (e.g., non-3GPP access). Note that as shown, the 5G MM may maintain individual connection management and registration management state machines for each connection. Additionally, a device (e.g., UE **106**) may register to a single PLMN (e.g., 5G CN) using 5G cellular access as well as non-cellular access. Further, it may be possible for the device to be in a connected state in one access and an idle state in another access and vice versa. Finally, there may be common 5G-MM procedures (e.g., registration, de-registration, identification, authentication, as so forth) for both accesses.

Note that in various embodiments, one or more of the above described functional entities of the 5G NAS and/or 5G AS may be configured to perform methods for 5G NR FR2 based communications beam management, e.g., as further described herein.

5G NR FR2 Beam Management

In current implementations, the air interface defined by 3GPP for 5G NR is subdivided into two frequency ranges. Frequency range 1 (FR1) includes frequencies below 6 GHz and frequency range 2 (FR2) includes frequencies above 24 GHz (e.g., millimeter wave (mmW) spectrum). In 3GPP Release 15, the minimum channel bandwidth defined for FR2 is 50 MHz and the maximum channel bandwidth defined for FR2 is 400 MHz. Due to the analog beam forming associated with FR2 communications, beam management differs as compared to FR1 communications. In current implementations of 3GPP Release 15 only basic beam management solutions are defined. For example, 3GPP Release 15 only supports a medium access control (MAC) control element (CE) for update of quasi co-location (QCL) (beam) information for one physical uplink control channel (PUCCH) per MAC-CE. Further, multi-panel support, which may be more important for FR2 communications as compared to FR1 communications (e.g., to ensure mmW coverage due to different device holding positions and/or nearby blocking objects), has not been optimized for efficient panel management. As yet another example, in current implementations, power control is linked to SRI (sounding reference signal (SRS) resource indicator) or other reference signals and configured via radio resource control (RRC) signaling, which may not be optimal for changing conditions that may impact FR2 communications more so than FR1 communications.

Embodiments described herein provide systems, methods, and mechanisms for a user equipment device (UE), such as UE **106**, to perform beam management for FR2 based communications. In some embodiments, one MAC CE may update a group of PUCCHs to reduce latency and/or signaling overhead. In some embodiments, the UE may use a MAC CE to update uplink (UL) power control parameters. In some embodiments, the UE may report a logical antenna panel identifier to a network.

In some embodiments, a PUCCH group identifier (group ID) may be assigned to each PUCCH resource, e.g., to allow a MAC CE to update QCL for a group of PUCCHs. In some embodiments, a MAC CE may use either a resource ID (e.g., to update a specific PUCCH) or a group ID (e.g., to update a group of PUCCHs). In some embodiments, a group ID for PUCCH resources may be configured via an RRC message(s), e.g., during PUCCH configuration. In some embodiments, the UE may signal a PUCCH resource ID with a PUCCH group ID. In some embodiments, a PUCCH group ID may replace a PUCCH resource ID within a MAC CE.

In some embodiments, in cases in which a PUCCH resource ID is not valid (e.g., invalid) and/or is a reserved value, a MAC CE may be interpreted as an update to PUCCH resources configured in a bandwidth part (BWP), e.g., as indicated by a BWP identifier (ID). In some embodiments, the reserved value may be useable to indicate all serving cells instead of (and/or in addition to) the BWP. Thus, such a configuration may allow for a multiple cell update with a single MAC CE. In some embodiments, in cases in which a PUCCH resource ID is not valid (e.g., invalid) and/or is a reserved value, a MAC CE may be interpreted as an update to PUCCH resources configured in a cell indicated by a service cell identifier (ID), e.g., included in the MAC CE.

In some embodiments, after a beam failure condition (BFR), a UE may use a beam identified during a candidate beam search period, e.g., prior to explicit reconfiguration (by a base station, such as gNB **604**) of a PUCCH QCL via RRC signal or a MAC CE. In some embodiments, a default PUCCH group ID may be defined and/or specified. In some embodiments, a PUCCH group with a smallest PUCCH group ID in a current BWP may be the default PUCCH group. In some embodiments, the default PUCCH group may include services that are the least delay tolerant, e.g., to minimize data interruptions due to beam failure. In some embodiments, upon detection of a beam failure recovery procedure, a PUCCH resource in the default PUCCH group may have their QCL (beam) configuration updated with a newly discovered beam.

In some embodiments, a UE, such as UE **106**, may activate and/or deactivate antenna panels for transmission, e.g., without input from a network. In some embodiments, a UE may report capabilities associated with a QCL update. For example, in some embodiments, the UE may report a number (e.g., a maximum number, in some embodiments) of supported transmission configuration indicator (TCI) states across component carriers (CCs) within a band. In some embodiments, for a PUCCH, the UE may report a number (e.g., a maximum number, in some embodiments) of PUCCH groups supported per BWP, within a cell, and/or across CCs within a band. In some embodiments, active PUCCH resources overlapping in the time domain may belong (or be associated with) a PUCCH group (e.g., automatically) for intra-band. In some embodiments, active PUCCH resources overlapping in the time domain may belong (or be associated with) a PUCCH group for inter-band within a band group.

In some embodiments, a PUCCH group may reduce a number of beams a UE may be required to prepare and/or track for uplink (UL) transmissions. For example, in some embodiments, during a UE measurement reporting procedure, the UE may report a logic antenna panel ID used for reception to the network. Note that the logical antenna panel ID may not be related to a physical location of an antenna panel. In some embodiments, during the measurement reporting procedure, the UE may assign a logical antenna

panel ID to a channel state information reference signal (CSI-RS) resource indicator (CRI, e.g., which may be used to indicate a preferred beam) and/or a synchronization signal/physical broadcast channel (SSB/PBCH) block resource indicator (SSBRI) (e.g., a resource index for channel state information reference signal (CSIRS) or synchronization signal block (SSB)). In some embodiments, during sounding reference signal (SRS) transmissions, the UE may report a logical antenna panel ID used for transmission of the SRS to the network. In some embodiments, such a report (or such reporting) may be performed separately from SRS transmissions, e.g., by mapping SRI to the logical antenna panel ID. In some embodiments, when a base station (e.g., such as gNB 604) configures a beam for the UE via a transmission control indication (TCI), e.g., for downlink beam selection, or sounding reference indication (SRI), e.g., for uplink beam selection, the logical antenna panel ID may be established during the measurement reporting procedure. In other words, when the base station configures the beam for the UE via TCI or SRI, the base station may not implicitly indicate the logical antenna panel ID.

For example, FIG. 9 illustrates signaling between a base station and a UE to establish a beam and associated logical antenna panel ID, according to some embodiments. As shown, gNB 904 (which may be a gNB 604) may transmit a CSI-RS/SSB 910 to UE 906 (which may be a UE 106), e.g., as part of a beam configuration procedure. UE 906 may receive the CSI-RS/SSB 910 and perform beam measurements, including determining an antenna panel to use for communications with gNB 604. UE 906 may assign a logical antenna panel ID to the determined antenna panel and transmit beam report 912 to gNB 904. The beam report 912 may include the logical antenna panel ID. Upon receipt of the beam report 912, gNB 904 may configure a beam and transmit beam indication 914 to UE 906. The beam indication 914 may include the logical antenna panel ID to use for communications.

In some embodiments, a UE may perform power management functions in which antenna panels may be switched into various power consumption modes, e.g., active and in use, active and not in use, and/or not active and not in use. The UE, in some embodiments, may balance power consumption with latency. For example, in some embodiments, switching to an active antenna panel that is not in use may incur a switching delay (or latency) between 14 and 48 symbols. However, an active antenna panel (even when not in use) may consume a relatively high amount of power as compared to a deactivated antenna panel. Additionally, in some embodiments, switching to a deactivated panel may incur a switching delay between 224 and 336 symbols (including activation of the antenna panel). Thus, the UE may have to balance switching latency with power consumption, at least in some embodiments. In some embodiments, a UE, such as UE 106, may provide (and/or update) a switching time needed for each antenna panel to the network (e.g., to a base station, such as gNB 604). In some embodiments, the update to the network may be periodical and/or event based (e.g., when an activation/usage state of an antenna panel changes). In some embodiments, such an update (or report) may be transmitted via a PUCCH, a MAC-CE, and/or via RRC signaling. In some embodiments, each antenna panel may be assigned a logical antenna panel ID and the UE may provide updates based on respective antenna panel IDs.

For example, FIG. 10 illustrates a multiple antenna panel UE switching panel states, according to some embodiments. As shown, a UE 1006 (which may be a UE 106) may include

antenna panels 1020, 1022, 1024, and 1026. At a first time (t1), antenna panel 1020 may be activated and in use by UE 1006 and may have an associated switching latency (delay) of 14 symbols. Additionally, UE 1006 may be operating in a power state such that antenna panels 1022, 1024, and 1026 may be activated. Further, antenna panels 1022, 1024, and 1026 may have an associated switching latency (or delay) of 28 symbols. However, at a second time (t2), while antenna panel 1020 may remain activated and in use by UE 1006 (and may have an associated switching latency (delay) of 14 symbols), the UE 1006 may be operating in a different power state as compared to the first time. Thus, antenna panel 1022 may remain activated (with an associated switching latency (delay) of 28 symbols), however, antenna panels 1024 and 1026 may be deactivated. Such a deactivation may cause an associated switching latency (delay) to increase for antenna panels 1024 and 1026 to 224 symbols (as compared to 28 symbols when activated).

In some embodiments, when a base station, such as gNB 604, cannot ensure a latency requirement requested by a UE, such as UE 106, the system may be configured to adapt in one or more ways. For example, in some embodiments, when a base station violates (and/or cannot ensure) a UE reported requirement for beam switching latency, the UE behavior may be left to implementation (e.g., the UE may determine which beam and/or antenna panel to use without network input). As another example, in some embodiments,

when a base station violates (and/or cannot ensure) a UE reported requirement for beam switching latency, the UE behavior may be pre-specified (e.g., standardized) (e.g., the UE may select a beam used for a control resource set (CORESET) with a lowest ID in a current active antenna panel and/or the UE may select a last beam used for reception and/or transmission in a current active antenna panel). As a further example, in some embodiments, when a base station violates (and/or cannot ensure) a UE reported requirement for beam switching latency, the base station may indicate a logical antenna panel ID explicitly for measurement purpose (e.g. A-CSI-RS, A-SRS). In such instances, there may be a restriction such that at least one beam quality for the indicated antenna panel has been reported within a time window. As a yet further example, in some embodiments, when a base station violates (and/or cannot ensure) a UE reported requirement for beam switching latency, the UE may select an antenna panel and report a logical antenna panel ID to the base station, e.g., after a beam failure and prior to the UE reporting a beam quality for an antenna panel.

In some embodiments, a pathloss measurement reference signal (RS) for SRS/PUCCH/PUSCH may be updated via a MAC CE. In such embodiments, prior to a base station (e.g., such as gNB 604) reconfiguring a pathloss measurement RS, the base station may confirm that a UE (such as UE 106) can measure the pathloss measurement RS at least once prior to transmission of a subsequent SRS/PUSCH/PUCCH. In some embodiments, if the UE cannot measure the pathloss measurement RS in time, the UE may continue to use a prior RS, e.g., at least until a measurement can be completed. For example, FIG. 11 illustrates a timeline for pathloss measurement RS reconfiguration, according to some embodiments. As shown, a UE, such as UE 106 may receive a reference signal (RS) 1120 for pathloss measurement. The UE may then receive an RS reconfiguration indication 1122 (e.g., via a MAC CE). Further, the new RS, RS 1124, may be received by the UE for measurement and, after a time, t1, SRS 1126 may be received by the UE. However, since t1 is less than a required measurement time for the UE to measure

RS 1124, the UE may continue to use RS 1120. Additionally, the UE may receive an RS reconfiguration indication 1128 (e.g., via a MAC CE) and a new RS, RS 1128, may be received by the UE for measurement. Further, after a time, t2, SRS 1132 may be received by the UE and, since t2 is greater than the required measurement time for the UE to measure RS 1130, the UE may complete the measurement of SRS 1132 and begin to use RS 1130.

FIG. 12 illustrates a block diagram of an example of a method for Fifth Generation New Radio (5G NR) frequency range two (FR2) based communications beam management, according to some embodiments. The method shown in FIG. 12 may be used in conjunction with any of the systems, methods, or devices shown in the Figures, among other devices. In various embodiments, some of the method elements shown may be performed concurrently, in a different order than shown, or may be omitted. Additional method elements may also be performed as desired. As shown, this method may operate as follows.

At 1202, a UE, such as UE 106, may configure, via communications with a base station, such as base station 102 and/or gNB 604, serving the UE, a group identifier (ID) for physical uplink control channel (PUCCH) resources within a group of PUCCHs. In some embodiments, configuring the group ID may include comprises exchanging radio resource control (RRC) messages with the base station, e.g., during a PUCCH configuration procedure.

At 1204, the UE may receive, from the base station serving the UE, an update to one or more PUCCH resources within the group of PUCCHs via a medium access control (MAC) control element (CE). In some embodiments, the MAC CE may include one of a resource ID indicating an update to a PUCCH resource or the group ID to update the group of PUCCH resources. In some embodiments, one of the resource ID, the group ID, a bandwidth part ID, and/or a service cell ID may indicate one of a reserved value or an invalid value. In some embodiments, when the reserved value or the invalid value is received, the UE may assume that a single beam updates resources, bandwidth parts, component carriers, and/or groups associated with the UE. In some embodiments, the UE may interpret the MAC CE as an update to PUCCH resources configured in a bandwidth part, an update to PUCCH resources configured in cells serving the UE, or an update to PUCCH resources configured in a cell indicated by a service cell identifier included in the MAC CE. In some embodiments the bandwidth part may be indicated via a bandwidth part identifier.

At 1206, the UE may update, based on the MAC-CE, the one or more PUCCH resources within the group of PUCCHs.

In some embodiments, a default PUCCH group ID may be defined and/or specified. In some embodiments, wherein a default PUCCH group may include at least one of a PUCCH group with a smallest PUCCH group ID in a current bandwidth part or a PUCCH group that includes services that are the least delay tolerant.

In some embodiments, the UE may detect a beam failure condition and detect, after detection of the beam failure condition, a new valid beam. In some embodiments, prior to a receiving an update from a network (e.g., the base station) indicating a new beam, the UE may switch to the new valid beam for uplink transmissions. In some embodiments, switching to the new valid beam for uplink transmissions may include switching to the new beam for one or more of a physical uplink control channel (PUCCH) transmission, a sounding reference signal (SRS) transmission, a physical uplink shared channel (PUSCH) transmission, and/or a

physical random-access channel (PRACH) transmission. In some embodiments, detecting the new valid beam may include the UE identifying a beam identified during a prior candidate beam search period. In some embodiments, the UE may update a quasi co-location configuration of PUCCH resource in a default PUCCH group using the new valid beam.

In some embodiments, the UE may activate or deactivate at least one antenna panel used for transmissions with the base station. In such embodiments, the UE may report, to the base station, capabilities associated with a quasi co-location (QCL) update. The capabilities may include at least one of, one or more of, and/or both of a number (e.g., a maximum number, in some embodiments) of supported transmission configuration indicator (TCI) states across component carriers (CCs) within a band or a number (e.g., a maximum number, in some embodiments) of PUCCH groups supported per BWP, within a cell, or across CCs within a band for a PUCCH. In some embodiments, active PUCCH resources from different component carriers overlapping in the time domain may be associated with a PUCCH group for intra-band carrier aggregation or inter-band carrier aggregation.

In some embodiments, the UE may assign, during a measurement reporting procedure, a logical antenna panel ID to a channel state information reference signal (CSI-RS) resource indicator (CRI) or a synchronization signal/physical broadcast channel (SSB/PBCH) block resource indicator (SSBRI). In such embodiments, the UE may report, during the measurement reporting procedure, the logical antenna panel ID used for transmission to and/or reception from the base station. In some embodiments, the CRI/SSBRI may indicate a preferred beam. In some embodiments, the CRI/SSBRI may be a resource index for channel state information reference signal (CSIRS) or synchronization signal block (SSB)). In some embodiments, the UE may report, during sounding reference signal (SRS) transmissions, a logical antenna panel ID used for SRS transmissions.

In some embodiments, the UE may provide, to the base station, an indication of an antenna panel switching time via one of a MAC-CE, radio resource control (RRC) signaling, or signaling via a PUCCH. In some embodiments, the indication may be on a per antenna panel basis. In some embodiments, antenna panels may be identified via logical antenna panel identifiers (IDs). In some embodiments, the indication may be provided periodically to the base station. In some embodiments, the indication may be provided a-periodically to the base station, e.g., when UE decides to change a panel power management state. In some embodiments, the indication may be provided to the base station based on an occurrence of an event. In such embodiments, the event may include at least one of activation of an antenna panel, deactivation of an antenna panel, and/or switching to or from an antenna panel for transmission/reception.

In some embodiments, the UE may report, to the base station, a minimum latency requirement for switching antenna panels. In some embodiments, in response to the base station violating the minimum latency requirement, the UE may determine an antenna panel to use without input from the base station. In some embodiments, in response to the base station violating the minimum latency requirement, the UE may determine an antenna panel to use based on a pre-specified configuration. In such embodiments, the pre-specified configuration may include one of the UE selecting a beam used for a control resource set (CORESET) with a lowest ID in a current active antenna panel and/or the UE selecting a last beam used for reception and/or transmission

in a current active antenna panel. In some embodiments, in response to the base station violating the minimum latency requirement, the UE may receive, from the base station, an indication of a logical antenna panel ID to use for measurements. In some embodiments, the report may be provided via one of a MAC-CE, radio resource control (RRC) signaling, or signaling via a PUCCH.

In some embodiments, the UE may receive, from the base station, an update to a pathloss measurement reference signal (RS) via a MAC CE. In some embodiments, when (if) the UE cannot (reliably) measure the pathloss measurement RS (e.g., at least once) prior to transmission of a subsequent sounding reference signal/physical uplink shared channel/PUCCH (SRS/PUSCH/PUCCH), the UE may continue to use pathloss estimate from a prior RS to determine the uplink open loop transmit power until pathloss can be (reliably) measured from the new pathloss measurement RS.

FIG. 13 illustrates a block diagram of another example of a method for Fifth Generation New Radio (5G NR) frequency range two (FR2) based communications beam management, according to some embodiments. The method shown in FIG. 13 may be used in conjunction with any of the systems, methods, or devices shown in the Figures, among other devices. In various embodiments, some of the method elements shown may be performed concurrently, in a different order than shown, or may be omitted. Additional method elements may also be performed as desired. As shown, this method may operate as follows.

At **1302**, a UE, such as UE **106**, may activate and/or deactivate at least one antenna panel used for communications with a base station, such as base station **102** and/or gNB **604**, serving the UE.

At **1304**, the UE may report, to the base station, capabilities associated with a quasi co-location (QCL) update. The capabilities may include at least one of, one or more of, and/or both of a number (e.g., a maximum number, in some embodiments) of supported transmission configuration indicator (TCI) states across component carriers (CCs) within a band or a number (e.g., a maximum number, in some embodiments) of PUCCH groups supported per BWP, within a cell, or across CCs within a band for a PUCCH. In some embodiments, active PUCCH resources from different component carriers overlapping in the time domain may be associated with a PUCCH group for intra-band carrier aggregation or inter-band carrier aggregation.

In some embodiments, the UE may configure, via communications with the base station serving the UE, a group identifier (ID) for physical uplink control channel (PUCCH) resources within a group of PUCCHs. In some embodiments, configuring the group ID may include comprises exchanging radio resource control (RRC) messages with the base station, e.g., during a PUCCH configuration procedure. Additionally, the UE may receive, from the base station serving the UE, an update to one or more PUCCH resources within the group of PUCCHs via a medium access control (MAC) control element (CE). In some embodiments, the MAC CE may include one of a resource ID indicating an update to a PUCCH resource or the group ID to update the group of PUCCH resources. In some embodiments, one of the resource ID, the group ID, a bandwidth part ID, and/or a service cell ID may indicate one of a reserved value or an invalid value. In some embodiments, when the reserved value or the invalid value is received, the UE may assume that a single beam updates resources, bandwidth parts, component carriers, and/or groups associated with the UE. In some embodiments, the UE may interpret the MAC CE as an update to PUCCH resources configured in a bandwidth

part, an update to PUCCH resources configured in cells serving the UE, or an update to PUCCH resources configured in a cell indicated by a service cell identifier included in the MAC CE. In some embodiments the bandwidth part may be indicated via a bandwidth part identifier. In some embodiments, the UE may update, based on the MAC-CE, the one or more PUCCH resources within the group of PUCCHs.

In some embodiments, a default PUCCH group ID may be defined and/or specified. In some embodiments, wherein a default PUCCH group may include at least one of a PUCCH group with a smallest PUCCH group ID in a current bandwidth part or a PUCCH group that includes services that are the least delay tolerant.

In some embodiments, the UE may detect a beam failure condition and detect, after detection of the beam failure condition, a new valid beam. In some embodiments, prior to a receiving an update from a network (e.g., the base station) indicating a new beam, the UE may switch to the new valid beam for uplink transmissions. In some embodiments, switching to the new valid beam for uplink transmissions may include switching to the new beam for one or more of a physical uplink control channel (PUCCH) transmission, a sounding reference signal (SRS) transmission, a physical uplink shared channel (PUSCH) transmission, and/or a physical random-access channel (PRACH) transmission. In some embodiments, detecting the new valid beam may include the UE identifying a beam identified during a prior candidate beam search period. In some embodiments, the UE may update a quasi co-location configuration of PUCCH resource in a default PUCCH group using the new valid beam. In some embodiments, the UE may update a quasi co-location configuration of PUCCH resource in a default PUCCH group using the new valid beam. In some embodiments, the UE may update a quasi co-location configuration of PUCCH resource in a default PUCCH group using the new valid beam.

In some embodiments, the UE may assign, during a measurement reporting procedure, a logical antenna panel ID to a channel state information reference signal (CSI-RS) resource indicator (CRI) or a synchronization signal/physical broadcast channel (SSB/PBCH) block resource indicator (SSBRI). In such embodiments, the UE may report, during the measurement reporting procedure, the logical antenna panel ID used for transmission to and/or reception from the base station. In some embodiments, the CRI/SSBRI may indicate a preferred beam. In some embodiments, the CRI/SSBRI may be a resource index for channel state information reference signal (CSIRS) or synchronization signal block (SSB)). In some embodiments, the UE may report, during sounding reference signal (SRS) transmissions, a logical antenna panel ID used for SRS transmissions.

In some embodiments, the UE may provide, to the base station, an indication of an antenna panel switching time via one of a MAC-CE, radio resource control (RRC) signaling, or signaling via a PUCCH. In some embodiments, the indication may be on a per antenna panel basis. In some embodiments, antenna panels may be identified via logical antenna panel identifiers (IDs). In some embodiments, the indication may be provided periodically to the base station. In some embodiments, the indication may be provided a-periodically to the base station, e.g., when UE decides to change a panel power management state. In some embodiments, the indication may be provided to the base station based on an occurrence of an event. In such embodiments, the event may include at least one of activation of an antenna

panel, deactivation of an antenna panel, and/or switching to or from an antenna panel for transmission/reception.

In some embodiments, the UE may report, to the base station, a minimum latency requirement for switching antenna panels. In some embodiments, in response to the base station violating the minimum latency requirement, the UE may determine an antenna panel to use without input from the base station. In some embodiments, in response to the base station violating the minimum latency requirement, the UE may determine an antenna panel to use based on a pre-specified configuration. In such embodiments, the pre-specified configuration may include one of the UE selecting a beam used for a control resource set (CORESET) with a lowest ID in a current active antenna panel and/or the UE selecting a last beam used for reception and/or transmission in a current active antenna panel. In some embodiments, in response to the base station violating the minimum latency requirement, the UE may receive, from the base station, an indication of a logical antenna panel ID to use for measurements. In some embodiments, the report may be provided via one of a MAC-CE, radio resource control (RRC) signaling, or signaling via a PUCCH.

In some embodiments, the UE may receive, from the base station, an update to a pathloss measurement reference signal (RS) via a MAC CE. In some embodiments, when (if) the UE cannot (reliably) measure the pathloss measurement RS (e.g., at least once) prior to transmission of a subsequent sounding reference signal/physical uplink shared channel/PUCCH (SRS/PUSCH/PUCCH), the UE may continue to use pathloss estimate from a prior RS to determine the uplink open loop transmit power until pathloss can be (reliably) measured from the new pathloss measurement RS.

FIG. 14 illustrates a block diagram of another example of a method for Fifth Generation New Radio (5G NR) frequency range two (FR2) based communications beam management, according to some embodiments. The method shown in FIG. 14 may be used in conjunction with any of the systems, methods, or devices shown in the Figures, among other devices. In various embodiments, some of the method elements shown may be performed concurrently, in a different order than shown, or may be omitted. Additional method elements may also be performed as desired. As shown, this method may operate as follows.

At 1402, a UE, such as UE 106, may assign, during a measurement reporting procedure with a base station, such as base station 102 and/or gNB 604, serving the UE, a logical antenna panel ID to a channel state information reference signal (CSI-RS) resource indicator (CRI) or a synchronization signal/physical broadcast channel (SSB/PBCH) block resource indicator (SSBRI).

At 1404, the UE may report, during the measurement reporting procedure, the logical antenna panel ID used for transmission to and/or reception from the base station. In some embodiments, the CRI/SSBRI may indicate a preferred beam. In some embodiments, the CRI/SSBRI may be a resource index for channel state information reference signal (CSIRS) or synchronization signal block (SSB)). In some embodiments, the UE may report, during sounding reference signal (SRS) transmissions, a logical antenna panel ID used for SRS transmissions.

In some embodiments, the UE may configure, via communications with the base station serving the UE, a group identifier (ID) for physical uplink control channel (PUCCH) resources within a group of PUCCHs. In some embodiments, configuring the group ID may include comprises exchanging radio resource control (RRC) messages with the base station, e.g., during a PUCCH configuration procedure.

Additionally, the UE may receive, from the base station serving the UE, an update to one or more PUCCH resources within the group of PUCCHs via a medium access control (MAC) control element (CE). In some embodiments, the MAC CE may include one of a resource ID indicating an update to a PUCCH resource or the group ID to update the group of PUCCH resources. In some embodiments, one of the resource ID, the group ID, a bandwidth part ID, and/or a service cell ID may indicate one of a reserved value or an invalid value. In some embodiments, when the reserved value or the invalid value is received, the UE may assume that a single beam updates resources, bandwidth parts, component carriers, and/or groups associated with the UE. In some embodiments, the UE may interpret the MAC CE as an update to PUCCH resources configured in a bandwidth part, an update to PUCCH resources configured in cells serving the UE, or an update to PUCCH resources configured in a cell indicated by a service cell identifier included in the MAC CE. In some embodiments the bandwidth part may be indicated via a bandwidth part identifier. In some embodiments, the UE may update, based on the MAC-CE, the one or more PUCCH resources within the group of PUCCHs.

In some embodiments, a default PUCCH group ID may be defined and/or specified. In some embodiments, wherein a default PUCCH group may include at least one of a PUCCH group with a smallest PUCCH group ID in a current bandwidth part or a PUCCH group that includes services that are the least delay tolerant.

In some embodiments, the UE may detect a beam failure condition and detect, after detection of the beam failure condition, a new valid beam. In some embodiments, prior to a receiving an update from a network (e.g., the base station) indicating a new beam, the UE may switch to the new valid beam for uplink transmissions. In some embodiments, switching to the new valid beam for uplink transmissions may include switching to the new beam for one or more of a physical uplink control channel (PUCCH) transmission, a sounding reference signal (SRS) transmission, a physical uplink shared channel (PUSCH) transmission, and/or a physical random-access channel (PRACH) transmission. In some embodiments, detecting the new valid beam may include the UE identifying a beam identified during a prior candidate beam search period. In some embodiments, the UE may update a quasi co-location configuration of PUCCH resource in a default PUCCH group using the new valid beam.

In some embodiments, the UE may activate or deactivate at least one antenna panel used for transmissions with the base station. In such embodiments, the UE may report, to the base station, capabilities associated with a quasi co-location (QCL) update. The capabilities may include at least one of, one or more of, and/or both of a number (e.g., a maximum number, in some embodiments) of supported transmission configuration indicator (TCI) states across component carriers (CCs) within a band or a number (e.g., a maximum number, in some embodiments) of PUCCH groups supported per BWP, within a cell, or across CCs within a band for a PUCCH. In some embodiments, active PUCCH resources from different component carriers overlapping in the time domain may be associated with a PUCCH group for intra-band carrier aggregation or inter-band carrier aggregation.

In some embodiments, the UE may provide, to the base station, an indication of an antenna panel switching time via one of a MAC-CE, radio resource control (RRC) signaling, or signaling via a PUCCH. In some embodiments, the

indication may be on a per antenna panel basis. In some embodiments, antenna panels may be identified via logical antenna panel identifiers (IDs). In some embodiments, the indication may be provided periodically to the base station. In some embodiments, the indication may be provided a-periodically to the base station, e.g., when UE decides to change a panel power management state. In some embodiments, the indication may be provided to the base station based on an occurrence of an event. In such embodiments, the event may include at least one of activation of an antenna panel, deactivation of an antenna panel, and/or switching to or from an antenna panel for transmission/reception.

In some embodiments, the UE may report, to the base station, a minimum latency requirement for switching antenna panels. In some embodiments, in response to the base station violating the minimum latency requirement, the UE may determine an antenna panel to use without input from the base station. In some embodiments, in response to the base station violating the minimum latency requirement, the UE may determine an antenna panel to use based on a pre-specified configuration. In such embodiments, the pre-specified configuration may include one of the UE selecting a beam used for a control resource set (CORESET) with a lowest ID in a current active antenna panel and/or the UE selecting a last beam used for reception and/or transmission in a current active antenna panel. In some embodiments, in response to the base station violating the minimum latency requirement, the UE may receive, from the base station, an indication of a logical antenna panel ID to use for measurements. In some embodiments, the report may be provided via one of a MAC-CE, radio resource control (RRC) signaling, or signaling via a PUCCH.

In some embodiments, the UE may receive, from the base station, an update to a pathloss measurement reference signal (RS) via a MAC CE. In some embodiments, when (if) the UE cannot (reliably) measure the pathloss measurement RS (e.g., at least once) prior to transmission of a subsequent sounding reference signal/physical uplink shared channel/PUCCH (SRS/PUSCH/PUCCH), the UE may continue to use pathloss estimate from a prior RS to determine the uplink open loop transmit power until pathloss can be (reliably) measured from the new pathloss measurement RS.

FIG. 15 illustrates a block diagram of another example of a method for Fifth Generation New Radio (5G NR) frequency range two (FR2) based communications beam management, according to some embodiments. The method shown in FIG. 15 may be used in conjunction with any of the systems, methods, or devices shown in the Figures, among other devices. In various embodiments, some of the method elements shown may be performed concurrently, in a different order than shown, or may be omitted. Additional method elements may also be performed as desired. As shown, this method may operate as follows.

At 1502, a UE, such as UE 106, may attach to a base station, such as base station 102 and/or gNB 604. In some embodiments, the attachment may include exchange of one or more configuration messages.

At 1504, the UE may provide, to the base station, an indication of an antenna panel switching time. In some embodiments, the indication may be provided via one of a MAC-CE, radio resource control (RRC) signaling, or signaling via a PUCCH.

In some embodiments, the indication may be on a per antenna panel basis. In some embodiments, antenna panels may be identified via logical antenna panel identifiers (IDs). In some embodiments, the indication may be provided periodically to the base station. In some embodiments, the

indication may be provided to the base station based on an occurrence of an event. In such embodiments, the event may include at least one of activation of an antenna panel, deactivation of an antenna panel, and/or switching to or from an antenna panel for transmission/reception.

In some embodiments, the UE may configure, via communications with the base station serving the UE, a group identifier (ID) for physical uplink control channel (PUCCH) resources within a group of PUCCHs. In some embodiments, configuring the group ID may include comprises exchanging radio resource control (RRC) messages with the base station, e.g., during a PUCCH configuration procedure. Additionally, the UE may receive, from the base station serving the UE, an update to one or more PUCCH resources within the group of PUCCHs via a medium access control (MAC) control element (CE). In some embodiments, the MAC CE may include one of a resource ID indicating an update to a PUCCH resource or the group ID to update the group of PUCCH resources. In some embodiments, one of the resource ID, the group ID, a bandwidth part ID, and/or a service cell ID may indicate one of a reserved value or an invalid value. In some embodiments, when the reserved value or the invalid value is received, the UE may assume that a single beam updates resources, bandwidth parts, component carriers, and/or groups associated with the UE. In some embodiments, the UE may interpret the MAC CE as an update to PUCCH resources configured in a bandwidth part, an update to PUCCH resources configured in cells serving the UE, or an update to PUCCH resources configured in a cell indicated by a service cell identifier included in the MAC CE. In some embodiments the bandwidth part may be indicated via a bandwidth part identifier. In some embodiments, the UE may update, based on the MAC-CE, the one or more PUCCH resources within the group of PUCCHs.

In some embodiments, a default PUCCH group ID may be defined and/or specified. In some embodiments, wherein a default PUCCH group may include at least one of a PUCCH group with a smallest PUCCH group ID in a current bandwidth part or a PUCCH group that includes services that are the least delay tolerant.

In some embodiments, the UE may detect a beam failure condition and detect, after detection of the beam failure condition, a new valid beam. In some embodiments, prior to a receiving an update from a network (e.g., the base station) indicating a new beam, the UE may switch to the new valid beam for uplink transmissions. In some embodiments, switching to the new valid beam for uplink transmissions may include switching to the new beam for one or more of a physical uplink control channel (PUCCH) transmission, a sounding reference signal (SRS) transmission, a physical uplink shared channel (PUSCH) transmission, and/or a physical random-access channel (PRACH) transmission. In some embodiments, detecting the new valid beam may include the UE identifying a beam identified during a prior candidate beam search period. In some embodiments, the UE may update a quasi co-location configuration of PUCCH resource in a default PUCCH group using the new valid beam.

In some embodiments, the UE may activate or deactivate at least one antenna panel used for transmissions with the base station. In such embodiments, the UE may report, to the base station, capabilities associated with a quasi co-location (QCL) update. The capabilities may include at least one of, one or more of, and/or both of a number (e.g., a maximum number, in some embodiments) of supported transmission configuration indicator (TCI) states across component

careers (CCs) within a band or a number (e.g., a maximum number, in some embodiments) of PUCCH groups supported per BWP, within a cell, or across CCs within a band for a PUCCH. In some embodiments, active PUCCH resources from different component carriers overlapping in the time domain may be associated with a PUCCH group for intra-band carrier aggregation or inter-band carrier aggregation.

In some embodiments, the UE may assign, during a measurement reporting procedure, a logical antenna panel ID to a channel state information reference signal (CSI-RS) resource indicator (CRI) or a synchronization signal/physical broadcast channel (SSB/PBCH) block resource indicator (SSBRI). In such embodiments, the UE may report, during the measurement reporting procedure, the logical antenna panel ID used for transmission to and/or reception from the base station. In some embodiments, the CRI/SSBRI may indicate a preferred beam. In some embodiments, the CRI/SSBRI may be a resource index for channel state information reference signal (CSIRS) or synchronization signal block (SSB)). In some embodiments, the UE may report, during sounding reference signal (SRS) transmissions, a logical antenna panel ID used for SRS transmissions.

In some embodiments, the UE may report, to the base station, a minimum latency requirement for switching antenna panels. In some embodiments, in response to the base station violating the minimum latency requirement, the UE may determine an antenna panel to use without input from the base station. In some embodiments, in response to the base station violating the minimum latency requirement, the UE may determine an antenna panel to use based on a pre-specified configuration. In such embodiments, the pre-specified configuration may include one of the UE selecting a beam used for a control resource set (CORESET) with a lowest ID in a current active antenna panel and/or the UE selecting a last beam used for reception and/or transmission in a current active antenna panel. In some embodiments, in response to the base station violating the minimum latency requirement, the UE may receive, from the base station, an indication of a logical antenna panel ID to use for measurements. In some embodiments, the report may be provided via one of a MAC-CE, radio resource control (RRC) signaling, or signaling via a PUCCH.

In some embodiments, the UE may receive, from the base station, an update to the pathloss measurement reference signal (RS) via a MAC CE. In some embodiments, when (if) the UE cannot (reliably) measure the pathloss measurement RS (e.g., at least once) prior to transmission of a subsequent sounding reference signal/physical uplink shared channel/PUCCH (SRS/PUSCH/PUCCH), the UE may continue to use pathloss estimate from a prior RS to determine the uplink open loop transmit power until pathloss can be (reliably) measured from the new pathloss measurement RS.

FIG. 16 illustrates a block diagram of another example of a method for Fifth Generation New Radio (5G NR) frequency range two (FR2) based communications beam management, according to some embodiments. The method shown in FIG. 16 may be used in conjunction with any of the systems, methods, or devices shown in the Figures, among other devices. In various embodiments, some of the method elements shown may be performed concurrently, in a different order than shown, or may be omitted. Additional method elements may also be performed as desired. As shown, this method may operate as follows.

At 1602, a UE, such as UE 106, may report, to the base station, such as base station 102 and/or gNB 604, serving the UE, a minimum latency requirement for switching antenna panels.

At 1604, the UE may, in response to the base station violating the minimum latency requirement, determine an antenna panel to use without input from the base station. In some embodiments, in response to the base station violating the minimum latency requirement, the UE may determine an antenna panel to use based on a pre-specified configuration. In such embodiments, the pre-specified configuration may include one of the UE selecting a beam used for a control resource set (CORESET) with a lowest ID in a current active antenna panel and/or the UE selecting a last beam used for reception and/or transmission in a current active antenna panel. In some embodiments, in response to the base station violating the minimum latency requirement, the UE may receive, from the base station, an indication of a logical antenna panel ID to use for measurements. In some embodiments, the report may be provided via one of a MAC-CE, radio resource control (RRC) signaling, or signaling via a PUCCH.

In some embodiments, the UE may configure, via communications with the base station serving the UE, a group identifier (ID) for physical uplink control channel (PUCCH) resources within a group of PUCCHs. In some embodiments, configuring the group ID may include comprises exchanging radio resource control (RRC) messages with the base station, e.g., during a PUCCH configuration procedure. Additionally, the UE may receive, from the base station serving the UE, an update to one or more PUCCH resources within the group of PUCCHs via a medium access control (MAC) control element (CE). In some embodiments, the MAC CE may include one of a resource ID indicating an update to a PUCCH resource or the group ID to update the group of PUCCH resources. In some embodiments, one of the resource ID, the group ID, a bandwidth part ID, and/or a service cell ID may indicate one of a reserved value or an invalid value. In some embodiments, when the reserved value or the invalid value is received, the UE may assume that a single beam updates resources, bandwidth parts, component carriers, and/or groups associated with the UE. In some embodiments, the UE may interpret the MAC CE as an update to PUCCH resources configured in a bandwidth part, an update to PUCCH resources configured in cells serving the UE, or an update to PUCCH resources configured in a cell indicated by a service cell identifier included in the MAC CE. In some embodiments the bandwidth part may be indicated via a bandwidth part identifier. In some embodiments, the UE may update, based on the MAC-CE, the one or more PUCCH resources within the group of PUCCHs.

In some embodiments, a default PUCCH group ID may be defined and/or specified. In some embodiments, wherein a default PUCCH group may include at least one of a PUCCH group with a smallest PUCCH group ID in a current bandwidth part or a PUCCH group that includes services that are the least delay tolerant.

In some embodiments, the UE may detect a beam failure condition and detect, after detection of the beam failure condition, a new valid beam. In some embodiments, prior to a receiving an update from a network (e.g., the base station) indicating a new beam, the UE may switch to the new valid beam for uplink transmissions. In some embodiments, switching to the new valid beam for uplink transmissions may include switching to the new beam for one or more of a physical uplink control channel (PUCCH) transmission, a

sounding reference signal (SRS) transmission, a physical uplink shared channel (PUSCH) transmission, and/or a physical random-access channel (PRACH) transmission. In some embodiments, detecting the new valid beam may include the UE identifying a beam identified during a prior candidate beam search period. In some embodiments, the UE may update a quasi co-location configuration of PUCCH resource in a default PUCCH group using the new valid beam.

In some embodiments, the UE may activate or deactivate at least one antenna panel used for transmissions with the base station. In such embodiments, the UE may report, to the base station, capabilities associated with a quasi co-location (QCL) update. The capabilities may include at least one of, one or more of, and/or both of a number (e.g., a maximum number, in some embodiments) of supported transmission configuration indicator (TCI) states across component carriers (CCs) within a band or a number (e.g., a maximum number, in some embodiments) of PUCCH groups supported per BWP, within a cell, or across CCs within a band for a PUCCH. In some embodiments, active PUCCH resources from different component carriers overlapping in the time domain may be associated with a PUCCH group for intra-band carrier aggregation or inter-band carrier aggregation.

In some embodiments, the UE may assign, during a measurement reporting procedure, a logical antenna panel ID to a channel state information reference signal (CSI-RS) resource indicator (CRI) or a synchronization signal/physical broadcast channel (SSB/PBCH) block resource indicator (SSBRI). In such embodiments, the UE may report, during the measurement reporting procedure, the logical antenna panel ID used for transmission to and/or reception from the base station. In some embodiments, the CRI/SSBRI may indicate a preferred beam. In some embodiments, the CRI/SSBRI may be a resource index for channel state information reference signal (CSI-RS) or synchronization signal block (SSB). In some embodiments, the UE may report, during sounding reference signal (SRS) transmissions, a logical antenna panel ID used for SRS transmissions.

In some embodiments, the UE may provide, to the base station, an indication of an antenna panel switching time via one of a MAC-CE, radio resource control (RRC) signaling, or signaling via a PUCCH. In some embodiments, the indication may be on a per antenna panel basis. In some embodiments, antenna panels may be identified via logical antenna panel identifiers (IDs). In some embodiments, the indication may be provided periodically to the base station. In some embodiments, the indication may be provided a-periodically to the base station, e.g., when UE decides to change a panel power management state. In some embodiments, the indication may be provided to the base station based on an occurrence of an event. In such embodiments, the event may include at least one of activation of an antenna panel, deactivation of an antenna panel, and/or switching to or from an antenna panel for transmission/reception.

In some embodiments, the UE may receive, from the base station, an update to a pathloss measurement reference signal (RS) via a MAC CE. In some embodiments, when (if) the UE cannot (reliably) measure the pathloss measurement RS (e.g., at least once) prior to transmission of a subsequent sounding reference signal/physical uplink shared channel/PUCCH (SRS/PUSCH/PUCCH), the UE may continue to use pathloss estimate from a prior RS to determine the uplink open loop transmit power until pathloss can be (reliably) measured from the new pathloss measurement RS.

FIG. 17 illustrates a block diagram of another example of a method for Fifth Generation New Radio (5G NR) frequency range two (FR2) based communications beam management, according to some embodiments. The method shown in FIG. 17 may be used in conjunction with any of the systems, methods, or devices shown in the Figures, among other devices. In various embodiments, some of the method elements shown may be performed concurrently, in a different order than shown, or may be omitted. Additional method elements may also be performed as desired. As shown, this method may operate as follows.

At 1702, a UE, such as UE 106, may attach to a base station, such as base station 102 and/or gNB 604. In some embodiments, the attachment may include exchange of one or more configuration messages.

At 1704, the UE may receive, from the base station, an update to a pathloss measurement reference signal (RS) via a MAC CE. In some embodiments, when (if) the UE cannot (reliably) measure the pathloss measurement RS (e.g., at least once) prior to transmission of a subsequent sounding reference signal/physical uplink shared channel/PUCCH (SRS/PUSCH/PUCCH), the UE may continue to use pathloss estimate from a prior RS to determine the uplink open loop transmit power until pathloss can be (reliably) measured from the new pathloss measurement RS.

In some embodiments, the UE may configure, via communications with the base station serving the UE, a group identifier (ID) for physical uplink control channel (PUCCH) resources within a group of PUCCHs. In some embodiments, configuring the group ID may include comprises exchanging radio resource control (RRC) messages with the base station, e.g., during a PUCCH configuration procedure. Additionally, the UE may receive, from the base station serving the UE, an update to one or more PUCCH resources within the group of PUCCHs via a medium access control (MAC) control element (CE). In some embodiments, the MAC CE may include one of a resource ID indicating an update to a PUCCH resource or the group ID to update the group of PUCCH resources. In some embodiments, one of the resource ID, the group ID, a bandwidth part ID, and/or a service cell ID may indicate one of a reserved value or an invalid value. In some embodiments, when the reserved value or the invalid value is received, the UE may assume that a single beam updates resources, bandwidth parts, component carriers, and/or groups associated with the UE. In some embodiments, the UE may interpret the MAC CE as an update to PUCCH resources configured in a bandwidth part, an update to PUCCH resources configured in cells serving the UE, or an update to PUCCH resources configured in a cell indicated by a service cell identifier included in the MAC CE. In some embodiments the bandwidth part may be indicated via a bandwidth part identifier. In some embodiments, the UE may update, based on the MAC-CE, the one or more PUCCH resources within the group of PUCCHs.

In some embodiments, a default PUCCH group ID may be defined and/or specified. In some embodiments, wherein a default PUCCH group may include at least one of a PUCCH group with a smallest PUCCH group ID in a current bandwidth part or a PUCCH group that includes services that are the least delay tolerant.

In some embodiments, the UE may detect a beam failure condition and detect, after detection of the beam failure condition, a new valid beam. In some embodiments, prior to a receiving an update from a network (e.g., the base station) indicating a new beam, the UE may switch to the new valid beam for uplink transmissions. In some embodiments,

switching to the new valid beam for uplink transmissions may include switching to the new beam for one or more of a physical uplink control channel (PUCCH) transmission, a sounding reference signal (SRS) transmission, a physical uplink shared channel (PUSCH) transmission, and/or a physical random-access channel (PRACH) transmission. In some embodiments, detecting the new valid beam may include the UE identifying a beam identified during a prior candidate beam search period. In some embodiments, the UE may update a quasi co-location configuration of PUCCH resource in a default PUCCH group using the new valid beam.

In some embodiments, the UE may activate or deactivate at least one antenna panel used for transmissions with the base station. In such embodiments, the UE may report, to the base station, capabilities associated with a quasi co-location (QCL) update. The capabilities may include at least one of, one or more of, and/or both of a number (e.g., a maximum number, in some embodiments) of supported transmission configuration indicator (TCI) states across component carriers (CCs) within a band or a number (e.g., a maximum number, in some embodiments) of PUCCH groups supported per BWP, within a cell, or across CCs within a band for a PUCCH. In some embodiments, active PUCCH resources from different component carriers overlapping in the time domain may be associated with a PUCCH group for intra-band carrier aggregation or inter-band carrier aggregation.

In some embodiments, the UE may assign, during a measurement reporting procedure, a logical antenna panel ID to a channel state information reference signal (CSI-RS) resource indicator (CRI) or a synchronization signal/physical broadcast channel (SSB/PBCH) block resource indicator (SSBRI). In such embodiments, the UE may report, during the measurement reporting procedure, the logical antenna panel ID used for transmission to and/or reception from the base station. In some embodiments, the CRI/SSBRI may indicate a preferred beam. In some embodiments, the CRI/SSBRI may be a resource index for channel state information reference signal (CSI-RS) or synchronization signal block (SSB)). In some embodiments, the UE may report, during sounding reference signal (SRS) transmissions, a logical antenna panel ID used for SRS transmissions.

In some embodiments, the UE may provide, to the base station, an indication of an antenna panel switching time via one of a MAC-CE, radio resource control (RRC) signaling, or signaling via a PUCCH. In some embodiments, the indication may be on a per antenna panel basis. In some embodiments, antenna panels may be identified via logical antenna panel identifiers (IDs). In some embodiments, the indication may be provided periodically to the base station. In some embodiments, the indication may be provided a-periodically to the base station, e.g., when UE decides to change a panel power management state. In some embodiments, the indication may be provided to the base station based on an occurrence of an event. In such embodiments, the event may include at least one of activation of an antenna panel, deactivation of an antenna panel, and/or switching to or from an antenna panel for transmission/reception.

In some embodiments, the UE may report, to the base station, a minimum latency requirement for switching antenna panels. In some embodiments, in response to the base station violating the minimum latency requirement, the UE may determine an antenna panel to use without input from the base station. In some embodiments, in response to the base station violating the minimum latency requirement, the UE may determine an antenna panel to use based on a

pre-specified configuration. In such embodiments, the pre-specified configuration may include one of the UE selecting a beam used for a control resource set (CORESET) with a lowest ID in a current active antenna panel and/or the UE selecting a last beam used for reception and/or transmission in a current active antenna panel. In some embodiments, in response to the base station violating the minimum latency requirement, the UE may receive, from the base station, an indication of a logical antenna panel ID to use for measurements. In some embodiments, the report may be provided via one of a MAC-CE, radio resource control (RRC) signaling, or signaling via a PUCCH.

FIG. 18 illustrates a block diagram of another example of a method for Fifth Generation New Radio (5G NR) frequency range two (FR2) based communications beam management, according to some embodiments. The method shown in FIG. 18 may be used in conjunction with any of the systems, methods, or devices shown in the Figures, among other devices. In various embodiments, some of the method elements shown may be performed concurrently, in a different order than shown, or may be omitted. Additional method elements may also be performed as desired. As shown, this method may operate as follows.

At **1802**, a UE, such as UE **106**, may detect a beam failure condition. In some embodiments, prior to detection of the beam failure conditions, the UE may attach to a base station, such as base station **102** and/or gNB **604**. In some embodiments, the attachment may include exchange of one or more configuration messages.

At **1804**, the UE may, responsive to (and/or after) detection of the beam failure condition, detect a new valid beam. In some embodiments, detecting the new valid beam may include the UE identifying a beam identified during a prior candidate beam search period. In some embodiments, the UE may update a quasi co-location configuration of PUCCH resource in a default PUCCH group using the new valid beam.

At **1806**, prior to a receiving an update from a network (e.g., the base station) indicating a new beam, the UE may switch to the new valid beam for uplink transmissions. In some embodiments, switching to the new valid beam for uplink transmissions may include switching to the new beam for one or more of a physical uplink control channel (PUCCH) transmission, a sounding reference signal (SRS) transmission, a physical uplink shared channel (PUSCH) transmission, and/or a physical random-access channel (PRACH) transmission.

In some embodiments, the UE may configure, via communications with the base station serving the UE, a group identifier (ID) for physical uplink control channel (PUCCH) resources within a group of PUCCHs. In some embodiments, configuring the group ID may include comprises exchanging radio resource control (RRC) messages with the base station, e.g., during a PUCCH configuration procedure. Additionally, the UE may receive, from the base station serving the UE, an update to one or more PUCCH resources within the group of PUCCHs via a medium access control (MAC) control element (CE). In some embodiments, the MAC CE may include one of a resource ID indicating an update to a PUCCH resource or the group ID to update the group of PUCCH resources. In some embodiments, one of the resource ID, the group ID, a bandwidth part ID, and/or a service cell ID may indicate one of a reserved value or an invalid value. In some embodiments, when the reserved value or the invalid value is received, the UE may assume that a single beam updates resources, bandwidth parts, component carriers, and/or groups associated with the UE.

In some embodiments, the UE may interpret the MAC CE as an update to PUCCH resources configured in a bandwidth part, an update to PUCCH resources configured in cells serving the UE, or an update to PUCCH resources configured in a cell indicated by a service cell identifier included in the MAC CE. In some embodiments the bandwidth part may be indicated via a bandwidth part identifier. In some embodiments, the UE may update, based on the MAC-CE, the one or more PUCCH resources within the group of PUCCHs.

In some embodiments, a default PUCCH group ID may be defined and/or specified. In some embodiments, wherein a default PUCCH group may include at least one of a PUCCH group with a smallest PUCCH group ID in a current bandwidth part or a PUCCH group that includes services that are the least delay tolerant.

In some embodiments, the UE may activate or deactivate at least one antenna panel used for transmissions with the base station. In such embodiments, the UE may report, to the base station, capabilities associated with a quasi co-location (QCL) update. The capabilities may include at least one of, one or more of, and/or both of a number (e.g., a maximum number, in some embodiments) of supported transmission configuration indicator (TCI) states across component carriers (CCs) within a band or a number (e.g., a maximum number, in some embodiments) of PUCCH groups supported per BWP, within a cell, or across CCs within a band for a PUCCH. In some embodiments, active PUCCH resources from different component carriers overlapping in the time domain may be associated with a PUCCH group for intra-band carrier aggregation or inter-band carrier aggregation.

In some embodiments, the UE may assign, during a measurement reporting procedure, a logical antenna panel ID to a channel state information reference signal (CSI-RS) resource indicator (CRI) or a synchronization signal/physical broadcast channel (SSB/PBCH) block resource indicator (SSBRI). In such embodiments, the UE may report, during the measurement reporting procedure, the logical antenna panel ID used for transmission to and/or reception from the base station. In some embodiments, the CRI/SSBRI may indicate a preferred beam. In some embodiments, the CRI/SSBRI may be a resource index for channel state information reference signal (CSIRS) or synchronization signal block (SSB)). In some embodiments, the UE may report, during sounding reference signal (SRS) transmissions, a logical antenna panel ID used for SRS transmissions.

In some embodiments, the UE may provide, to the base station, an indication of an antenna panel switching time via one of a MAC-CE, radio resource control (RRC) signaling, or signaling via a PUCCH. In some embodiments, the indication may be on a per antenna panel basis. In some embodiments, antenna panels may be identified via logical antenna panel identifiers (IDs). In some embodiments, the indication may be provided periodically to the base station. In some embodiments, the indication may be provided a-periodically to the base station, e.g., when UE decides to change a panel power management state. In some embodiments, the indication may be provided to the base station based on an occurrence of an event. In such embodiments, the event may include at least one of activation of an antenna panel, deactivation of an antenna panel, and/or switching to or from an antenna panel for transmission/reception.

In some embodiments, the UE may report, to the base station, a minimum latency requirement for switching antenna panels. In some embodiments, in response to the base station violating the minimum latency requirement, the

UE may determine an antenna panel to use without input from the base station. In some embodiments, in response to the base station violating the minimum latency requirement, the UE may determine an antenna panel to use based on a pre-specified configuration. In such embodiments, the pre-specified configuration may include one of the UE selecting a beam used for a control resource set (CORESET) with a lowest ID in a current active antenna panel and/or the UE selecting a last beam used for reception and/or transmission in a current active antenna panel. In some embodiments, in response to the base station violating the minimum latency requirement, the UE may receive, from the base station, an indication of a logical antenna panel ID to use for measurements. In some embodiments, the report may be provided via one of a MAC-CE, radio resource control (RRC) signaling, or signaling via a PUCCH.

In some embodiments, the UE may receive, from the base station, an update to a pathloss measurement reference signal (RS) via a MAC CE. In some embodiments, when (if) the UE cannot (reliably) measure the pathloss measurement RS (e.g., at least once) prior to transmission of a subsequent sounding reference signal/physical uplink shared channel/PUCCH (SRS/PUSCH/PUCCH), the UE may continue to use pathloss estimate from a prior RS to determine the uplink open loop transmit power until pathloss can be (reliably) measured from the new pathloss measurement RS.

It is well understood that the use of personally identifiable information should follow privacy policies and practices that are generally recognized as meeting or exceeding industry or governmental requirements for maintaining the privacy of users. In particular, personally identifiable information data should be managed and handled so as to minimize risks of unintentional or unauthorized access or use, and the nature of authorized use should be clearly indicated to users.

Embodiments of the present disclosure may be realized in any of various forms. For example, some embodiments may be realized as a computer-implemented method, a computer-readable memory medium, or a computer system. Other embodiments may be realized using one or more custom-designed hardware devices such as ASICs. Still other embodiments may be realized using one or more programmable hardware elements such as FPGAs.

In some embodiments, a non-transitory computer-readable memory medium may be configured so that it stores program instructions and/or data, where the program instructions, if executed by a computer system, cause the computer system to perform a method, e.g., any of the method embodiments described herein, or, any combination of the method embodiments described herein, or, any subset of any of the method embodiments described herein, or, any combination of such subsets.

In some embodiments, a device (e.g., a UE 106) may be configured to include a processor (or a set of processors) and a memory medium, where the memory medium stores program instructions, where the processor is configured to read and execute the program instructions from the memory medium, where the program instructions are executable to implement any of the various method embodiments described herein (or, any combination of the method embodiments described herein, or, any subset of any of the method embodiments described herein, or, any combination of such subsets). The device may be realized in any of various forms.

Although the embodiments above have been described in considerable detail, numerous variations and modifications will become apparent to those skilled in the art once the

37

above disclosure is fully appreciated. It is intended that the following claims be interpreted to embrace all such variations and modifications.

What is claimed is:

1. A method for frequency range two (FR2) beam management, comprising:
 - configuring, via communications with a base station, a group identifier (ID) for physical uplink control channel (PUCCH) resources within a group of PUCCHs;
 - updating, based on a medium access control (MAC) control element (CE) received from the base station, one or more PUCCH resources within the group of PUCCHs, wherein the MAC CE includes one of a resource ID or the group ID;
 - receiving, from the base station, an update to a pathloss measurement reference signal (RS); and
 - continuing, when the updated pathloss measurement RS cannot be measured at least once prior to transmission of a subsequent signal, to use a pathloss estimate from a prior RS to determine an uplink open loop transmit power until a pathloss can be measured from the updated pathloss measurement RS.
2. The method of claim 1, wherein the subsequent signal comprises at least one of:
 - a sounding reference signal;
 - a physical uplink shared channel; or
 - a physical uplink control channel.
3. The method of claim 1, further comprising:
 - attaching to the base station, wherein attaching to the base station comprises exchanging one or more configuration messages with the base station.
4. The method of claim 1, wherein the update to the pathloss measurement RS is received via a reconfiguration indication.
5. The method of claim 4, wherein the reconfiguration indication comprises a medium access control (MAC) control element (CE).
6. The method of claim 1, wherein configuring the group ID comprises exchanging radio resource control (RRC) messages during a PUCCH configuration procedure.
7. An apparatus, comprising:
 - a memory; and
 - at least one processor in communication with the memory and configured to:
 - configure, via communications with a base station, a group identifier (ID) for physical uplink control channel (PUCCH) resources within a group of PUCCHs;
 - update, based on a medium access control (MAC) control element (CE) received from the base station, one or more PUCCH resources within the group of PUCCHs, wherein the MAC CE includes one of a resource ID or the group ID;
 - receive, from the base station, an update to a pathloss measurement reference signal (RS); and
 - continue, when the updated pathloss measurement RS cannot be measured at least once prior to transmission of a subsequent signal, to use a pathloss estimate from a prior RS to determine an uplink open loop transmit power until a pathloss can be measured from the updated pathloss measurement RS.
8. The apparatus of claim 7, wherein the subsequent signal comprises at least one of:
 - a sounding reference signal;
 - a physical uplink shared channel; or
 - a physical uplink control channel.

38

9. The apparatus of claim 7, wherein the at least one processor is further configured to generate instructions to attach to the base station, and wherein to attach to the base station, the at least one processor is further configured to generate instructions to exchange one or more configuration messages with the base station.
10. The apparatus of claim 7, wherein the update to the pathloss measurement RS is received via a reconfiguration indication.
11. The apparatus of claim 10, wherein the reconfiguration indication comprises a medium access control (MAC) control element (CE).
12. The apparatus of claim 7, wherein to configure the group ID, the at least one processor is further configured to generate instructions to exchange radio resource control (RRC) messages during a PUCCH configuration procedure.
13. A user equipment device (UE), comprising:
 - at least one antenna;
 - at least one radio, wherein the at least one radio is configured to perform cellular communication using at least one radio access technology (RAT); and
 - one or more processors coupled to the at least one radio, wherein the one or more processors and the at least one radio are configured to perform voice and/or data communications;
 wherein the one or more processors are configured to cause the UE to:
 - configure, via communications with a base station, a group identifier (ID) for physical uplink control channel (PUCCH) resources within a group of PUCCHs;
 - update, based on a medium access control (MAC) control element (CE) received from the base station, one or more PUCCH resources within the group of PUCCHs, wherein the MAC CE includes one of a resource ID or the group ID;
 - receive, from the base station, an update to a pathloss measurement reference signal (RS); and
 - continue, when the updated pathloss measurement RS cannot be measured at least once prior to transmission of a subsequent signal, to use a pathloss estimate from a prior RS to determine an uplink open loop transmit power until a pathloss can be measured from the updated pathloss measurement RS.
14. The UE of claim 13, wherein the subsequent signal comprises at least one of:
 - a sounding reference signal;
 - a physical uplink shared channel; or
 - a physical uplink control channel.
15. The UE of claim 13, wherein the one or more processors are further configured to generate instructions to attach to the base station, and wherein to attach to the base station, the one or more processors are further configured to cause the UE to exchange one or more configuration messages.
16. The UE of claim 13, wherein the update to the pathloss measurement RS is received via a reconfiguration indication.
17. The UE of claim 16, wherein the reconfiguration indication comprises a medium access control (MAC) control element (CE).
18. The method of claim 1, further comprising:
 - reporting, to the base station, a minimum latency requirement for switching antenna panels; and

in response to the base station violating the minimum latency requirement, determining an antenna panel to use without input from the base station.

19. The method of claim 1, further comprising:
reporting, to the base station, a minimum latency require- 5
ment for switching antenna panels; and
in response to the base station violating the minimum latency requirement, determining an antenna panel to use based on a pre-specified configuration.

20. The method of claim 19, 10
wherein the pre-specified configuration includes one of
selecting a beam used for a control resource set (CORESET) with a lowest ID in a current active antenna panel or selecting a last beam used for reception or transmission in a current active antenna panel. 15

* * * * *